Department of Information & Communication Technology

Islamic University, Kushtia-7003

Lab Report
on
Signals and Systems Laboratory

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ICT 19-20

This Lab Report is submitted for the partial requirement of

ICT-2108 (Signals and Systems Laboratory)

Date of Submission: 26-09-2022

Submitted by	Submitted to
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EXPERIMENT NO: 01

NAME OF THE EXPERIMENT: Generation of basic signals — (i) Sinusoidal signals (ii) Exponential signals (iii) Unit impulse signals (iv) Unit step signals (v) Unit Ramp signals.

AIM:

(i) To represent the basic signals using MATLAB.
(ii) To generate and plot different types of signal

Objectives:

After completing this experiment, we are able to know about this:

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(i) Understand the representation of continous and discrete time signals in time domain.
(ii) Analyze continous and discrete-time signals and system in time domain
(iii) Basic knowledge of sinusoidal signals, exponential signals, unit impluse signals, unit step signals, unit ramp signals.
(iv) Implement the course concepts on using MATLAB.

THEORY:

Sinusoidals Signals:

A sinusoidal function or sinusoidal signal is a function that describes a smooth periodic oscillations.

Continuous Time Sinusoidal Signals:

A sinusoidal signal which is defined for every instant of time is called continous-time sinusoidal signal. The continuous time sinusoidal signal is given as follows ——

$$x(t) = A \sin(\omega t + \phi) = A \sin(2\pi f t + \phi)$$

Where,

A = A is the amplitude of the signal. That is the peak deviation of the signal from zero

$\omega = 2\pi f$ is the angular frequency in radians per second

$f = f$ is the frequency in radians per seconds.

ϕ = the phase angle in radians.

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All the continuous time signals are periodic signals. The time period (T) of a continuous time sinusoidal signal is given by,

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

The graphical representation or waveform of a continous time sinusoidal signal x(t) is shown in below

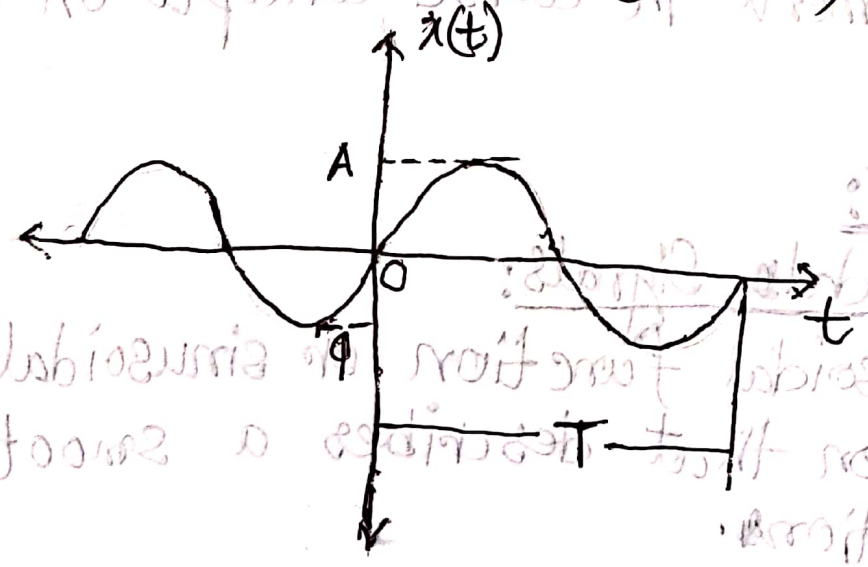

Figure: Continuous time sinusoidal signal

Discrete Time Sinusoidal Signals:

A sinusoidal signal which is defined only at discrete instants of time is called discrete-time sinusoidal signal. The discrete time sinusoidal signal is given as follows —

$$x[n] = A\cos(\omega n + \phi) = A\cos(2\pi f n + \phi)$$

Where, n is a integer.

The time period of the discrete-time sinusoidal signal is given by,

$$N = \frac{2\pi}{\omega} m$$, where N and m are integers.

The discrete time sinusoidal sequences may or may not periodic depending on the values of angular frequency(ω). For a discrete-time sinusoidal signal to be periodic, the angular frequency (ω) must be a rational multiple of 2π.

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The graphical representation of a discrete-time sinusoidal signal $x(n)$ is shown is below:

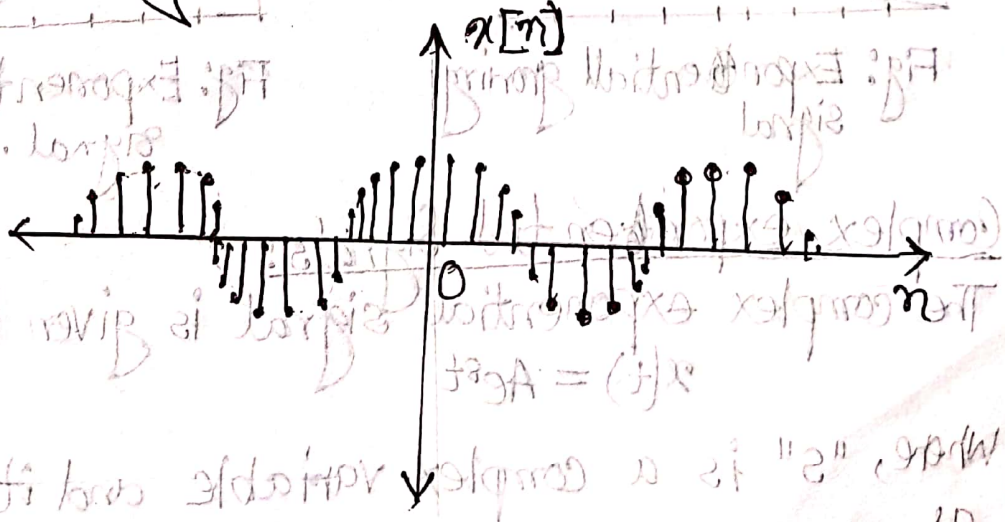

Figure: Discrete time sinusoidal signals

Exponential signals

Exponential signal is of two types. These two types of signal are real exponential signal and complex exponential signal which are given below.

Real Exponential Signals:

A real exponential signal is defined as

$$x(t) = Ae^{\alpha t}$$

Where both "A" and "α" are real. Depending on the value of "α" the signals will be different. If "α" is positive the signal $x(t)$ is a growing exponential and if "α" is negative then the signal $x(t)$ is a decaying exponential. For $\alpha = 0$, signal $x(t)$ will be constant.

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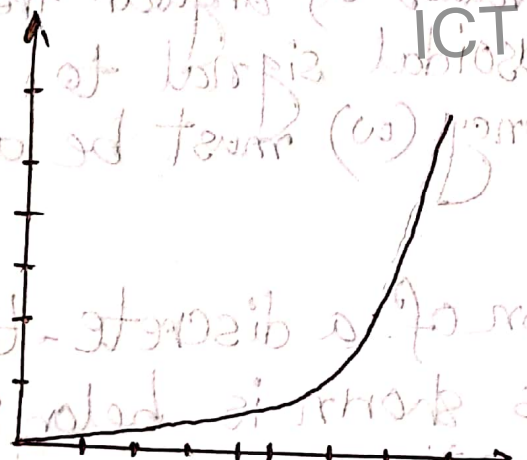

Fig: Exponentiall growing signal

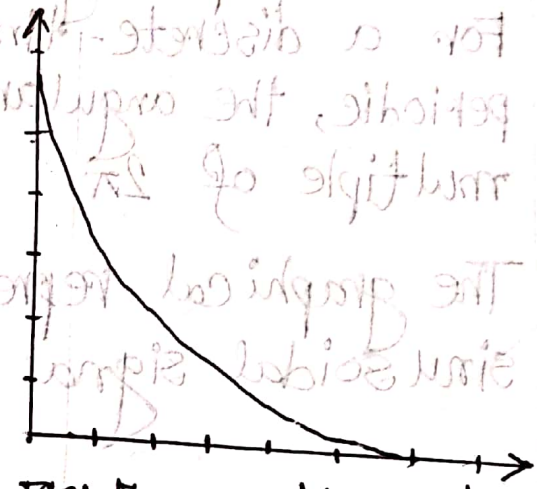

Fig: Exponentially decaying signal.

Complex exponential Signals:

The complex exponential signal is given by

$$x(t) = Ae^{st}$$

Where, "s" is a complex variable and it is defined as

$$s = a + jb$$

Therefore,

$$x(t) = e^{st} = e^{(a+jb)t} = e^{at} \cdot e^{jbt} \quad \text{---(i)}$$

Using Eulers identity,

$$e^{jbt} = e^{j\theta} = \cos\theta + j\sin\theta \quad \text{---(ii)}$$

Substituting Eq(2) in Eq(1), we have,

$$x(t) = e^{at}(\cos\theta + j\sin\theta)$$

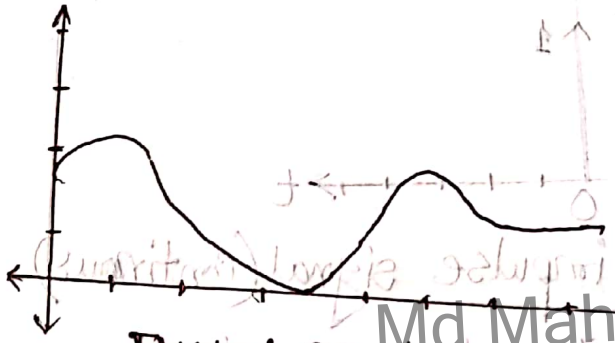

Figure: Complex Exponential signals

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Unit Impulse signal:

An ideal impulse signal is a signal that is zero everywhere but at the origin ($t=0$), it is infinitely high. Although, the area of the impulse is finite. The unit impulse signal is the most widely used standard signal used in the analysis of signals and systems.

Continuous-Time Unit Impulse Signal:

The continuous-time unit impluse signal is denoted by $\delta(t)$ and is defined as—

$$\delta(t) = \begin{cases} 1 & \text{for } t=0 \\ 0 & \text{for } t \neq 0 \end{cases}$$

Hence, by the definition, the unit impulse signal has zero amplitude everywhere except at $t=0$.

At the origin (t=0) the amplitude of impulse signal is infinity so that the drea under the curve is unity. The continuous time impulse signal is also called Dirac Delta signal.

The graphical representation of continuous-time unit impulse signal $\delta(t)$ is shown in below:

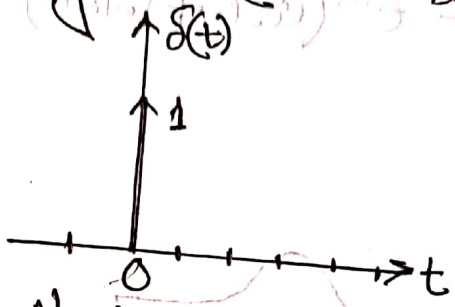

Figure: Unit impulse signal (Continous)

Discrete-Time Unit Impulse Signal:

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The discrete-time unit impulse signal is denoted by $\delta[n]$ and is defined as

$$\delta[n] = \begin{cases} 1 & \text{for } n=0 \\ 0 & \text{for } n \neq 0 \end{cases}$$

The discrete-time signal is also called unit sample sequence. The graphical representation of a discreate-time signal or unit sample sequence is shown in below:

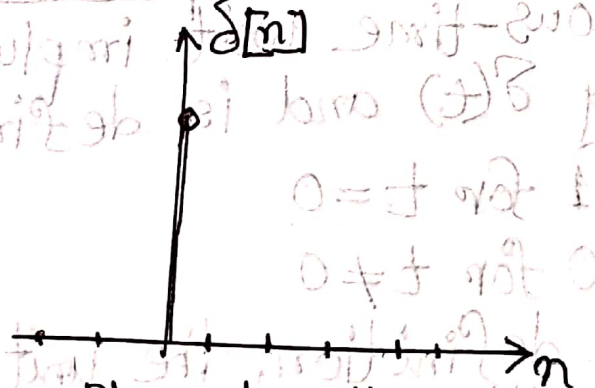

Figure: Discrete time unit impulse Signals.

Unit Step Signal

The step signal or step function is that type of standard signal which exists only for positive time and it is zero for negative time. In other words, a signal $x(t)$ is said to be step signal if and only if it exists for $t>0$ and zero for $t<0$. The step signal is an important signal used for analysis of many systems.

If a step signal is equivalent to applying a signal to a system whose magnitude suddenly changes and remains constant forever after application. If we want to obtain a signal which start at $t=0$, so that it may have a value of zero for $t<0$, then we only need to multiply the given signal with the unit step signal $u(t$

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Continuous-Time Unit Step Signals:

The unit step signal which is defined for every instant of time is known as continuous-time unit step signal. The continuous time unit step signal is denoted by $u(t)$.

Mathematically, the continuous time unit step signal $u(t)$ is defined as follows —

$$u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

From the above definition of the continuous time unit step function, it is clear that the unit

step function is zero, when the time is less than zero and when the time (t) is greater than or equal to zero, then u(t) is unity.

The graphical representation of the continuous time step unit function, it is shown in below

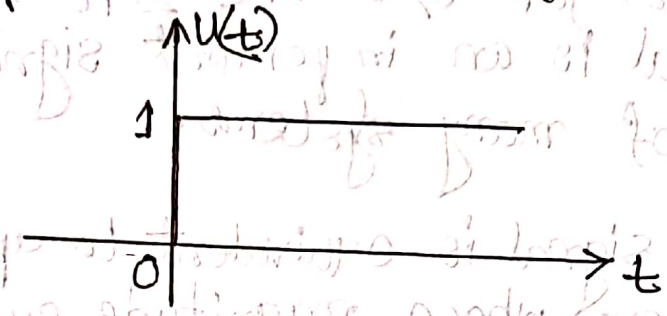

Figure: Continuous time unit step signal

Discrete time Unit Step signal

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The unit step signal which is defined only at discrete instant of time is known as discrete time unit step signal. It is denoted by u[n]. Mathematically, the discrete-time unit step signal or sequence u[n] is defined as follows:

$$u[n] = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$

The graphical representation of the discrete time unit step signal u[n] is shown in below:

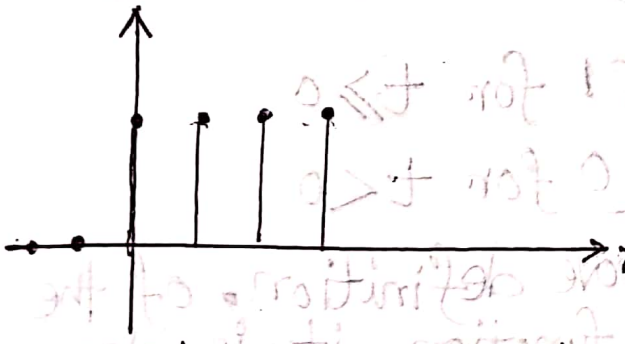

Figure: Discrete time unit step signal

Unit Ramp signal

A ramp function or ramp signal is a type of standard signal which starts at $t=0$ and increases linearly with time. The unit ramp function has unit slop.

Continuous-Time Unit Ramp Signal

The continuous time unit ramp signal is that function which starts at $t=0-$ and increases linearly with time. It is denoted by $r(t)$. Mathematically, the continuous-time unit ramp signal is defined as follows —

$$r(t) = \begin{cases} t & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

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Also,

$$r(t) = t \cdot u(t)$$

From the above equation, it is clear that the ramp signal is a signal whose magnitude varies linearly. The graphical representation of the continuous-time unit ramp signal is shown in below:

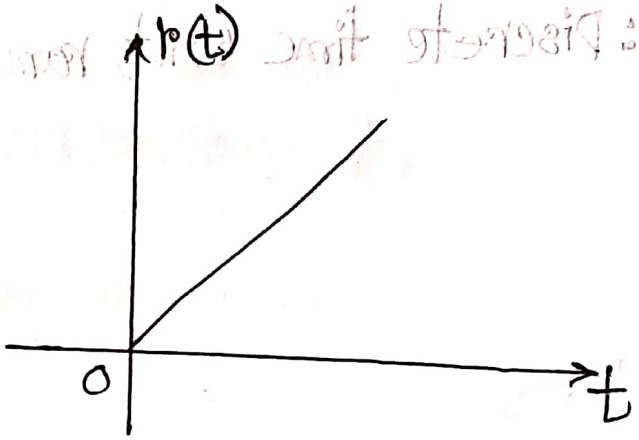

Figure: Continuous time unit ramp signal.

Discrete time Unit Ramp signal

The discrete time unit ramp signal is that fun which starts from $n=0$ and increases linear It is denoted by $r[n]$. It is signal whose amplitude varies linearly with time n. Mathematically, the discrete time unit ramp sequen is defined as —

$$r[n] = \begin{cases} n & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$

Or,

$$r[n] = n \cdot u[n]$$

The graphical representation of a discrete-time unit ramp sequence is shown in below:

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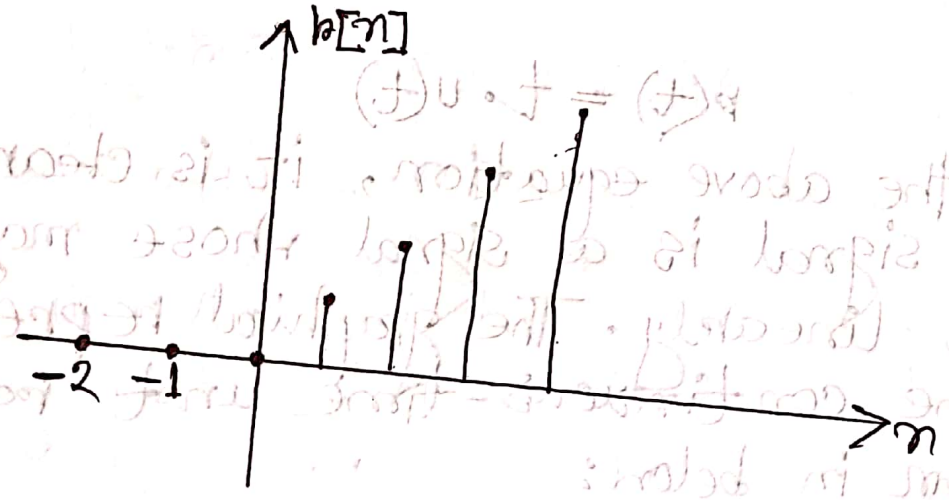

Figure: Discrete time unit ramp signal.

Calculation:

For sinusoidal signals,

Here, a Continuous time sinusoidal signals functions is—

$x(t) = 3\cos(3\pi t + \pi/3)$ and

Discrete time sinusoidal signal function is—

$x[n] = 3\cos(3\pi n + \pi/3)$

Where,

$A = 3;$

$\omega = 3\pi;$

$\phi = \pi/3;$

$T = 2\pi/\omega;$

```
t = 0:0.01:4*T;
n = 0:0.05:4*T;
```

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Real Exponential Signals

Here, a Real Exponential signals function is—

$x(t) = 3e^{0.8t}$

Where,

```
A = 3;
t = 0:0.1:3;
x = A*exp(0.8*t);
```

For Complex Exponential Signal:

Here, a complex exponential function is —

$x(t) = e^{(-0.1+0.3j)t}$

Where,

```
t = -20:20;
a = a Complex variable = a+jb
  = -0.1+0.3j  [a=-0.1 , b=0.3]
x = exp(a*t)
plot(t, imag(x));
```

Algorithm

For sinusoidal Continuous & Discrete time signal,

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(i) Start

(ii) Define the continuous and discrete sinusoidal function $x(t)$ or $x[n]$.

(iii) Get the value of t or n.

(iv) Plotting the value of the function and t or n into the x or y axis.

(v) Stop.

For exponential singal (Real & Complex),

(i) start

(ii) Define the ~~continuous and discrete~~ Real and Complex exponential signal function $x(t)$.

(iii) Get the value of t;

(iv) Plotting the value of the function with respect to t.

(v) Stop.

For unit step signal (Continuous & Discrete),

(i) Start
(ii) Define the logic [t>=0] = u
(iii) Get the value of the range of t.
(iv) Plot the value of the function and t.
(v) Stop.

For unit impulse signal (Continuous & Discrete),

(i) Start
(ii) Define the logic [t==0] = x
(iii) Get the value of the range of t.
(iv) Plot the value of the function and t
(v) Stop.

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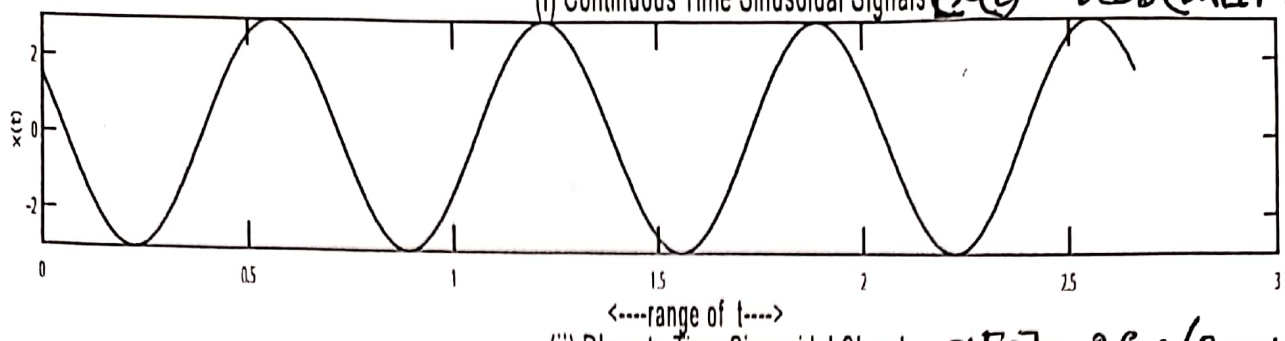
For unit ramp signal (Continuous & Discrete),

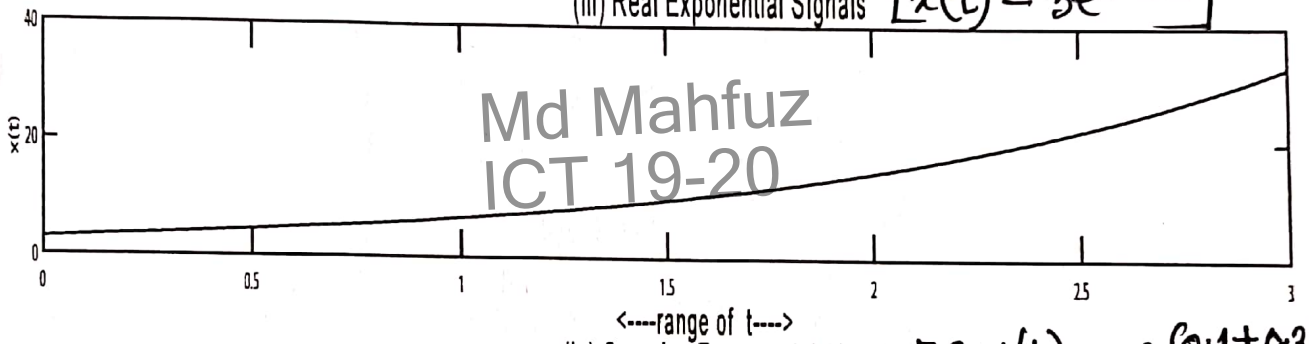
(i) Start
(ii) Define the logic [t*(t>=0)] = U(r)
(iii) Get the value of the range of t.
(iv) Plot the value of the function with t.
(v) Stop.

Output:

(i) Continuous Time Sinusoidal Signals $[x(t) = 3\cos(3\pi t + \frac{\pi}{3})]$



(iii) Real Exponential Signals $[x(t) = 3e^{0.6t}]$



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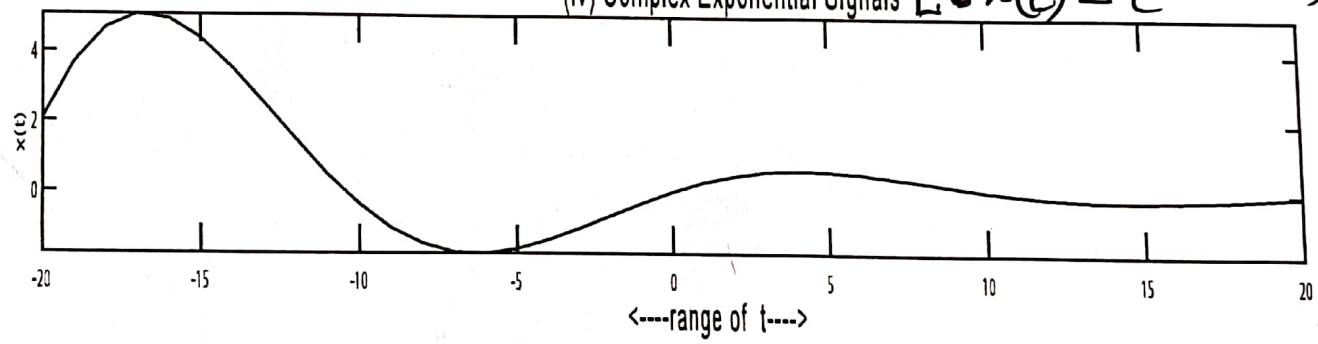
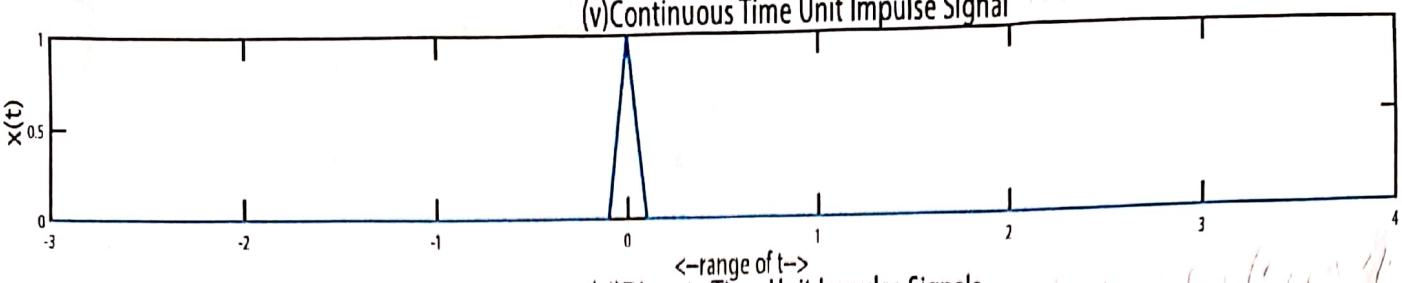
(iv) Complex Exponential Signals $[x(t) = e^{(0.1+0.3j)t}]$



(v)Continuous Time Unit Impulse Signal



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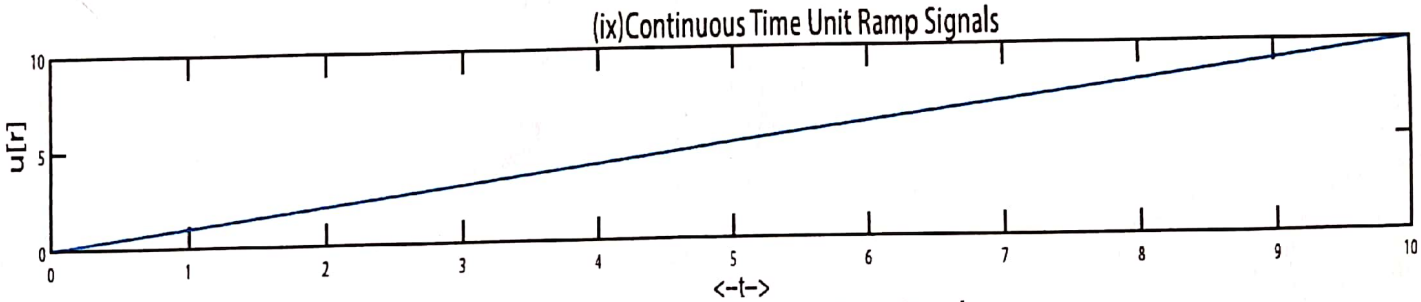
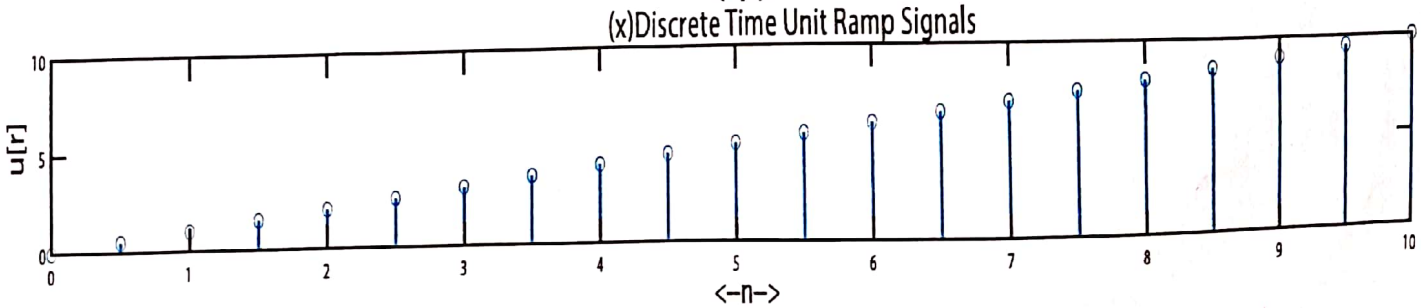
(ix)Continuous Time Unit Ramp Signals



(x)Discrete Time Unit Ramp Signals



Result and Discussion:

(i) A sine wave or sinusoidal wave is the most natural representation of how many things in nature change state. A sine wave shows how amplitude of the variables changes with time.

(ii) Real Exponential Signal are decaying or increasing signal.

(iii) Complex Exponential Signals have different amplitude.

(iv) A Ramp signal increases linearly with time.

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Experiment No: 2

Name of the Experiment: Basic Operation on Signals — (i) Addition (ii) Subtraction (iii) Multiplication (iv) Even and Odd signals (v) Amplitude scaling (vi) Time scaling (vii) Time reversal etc.

AIM:

To represent operations performed on dependent variables and independent variable.

Objectives:

After complete this experiment, we are able to know about this:-

(i) How to work two or more signals.
(ii) To compress or extent a signal.
(iii) Know about the even and odd signals.
(iv) How to performed on dependent and independent variables.
(v) Implement the course concept on using MATLAB.

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Theory:

Operation performed on dependent variables:

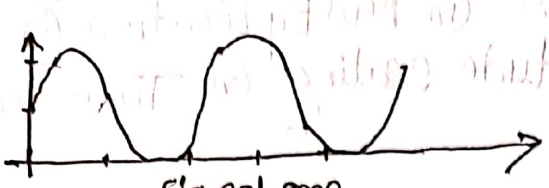
Addition:

Let $x_1(t)$ and $x_2(t)$ denote a pair of continous time signals. This signal $y(t)$ obtained by the addition of $x_1(t)$ and $x_2(t)$ is defined by

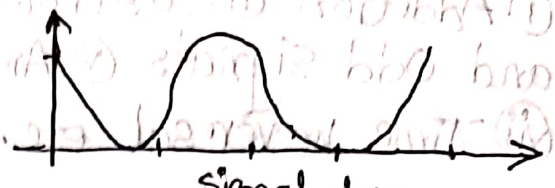
$$y(t) = x_1(t) + x_2(t) \quad \text{——(i)}$$

For discrete-time signals, we write;

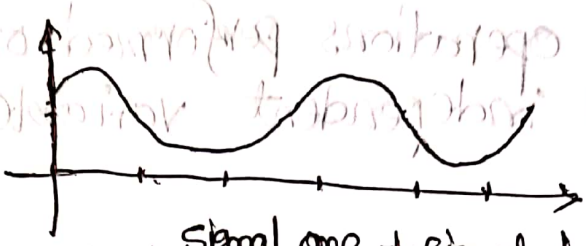
$$y[n] = x_1[n] + x_2[n]$$



signal one



signal two



signal one + signal two (Addition)

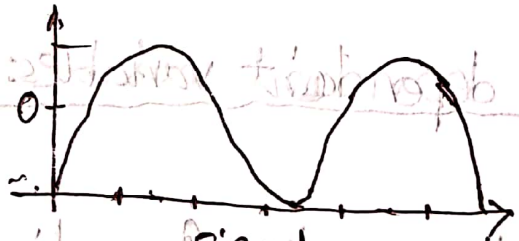
Subtraction:

Let $x_1(t)$ and $x_2(t)$ denote a pair of continuous time signals. The signal $y(t)$ obtained by the ~~addition~~ subtraction of $x_1(t)$ and $x_2(t)$ is defined by

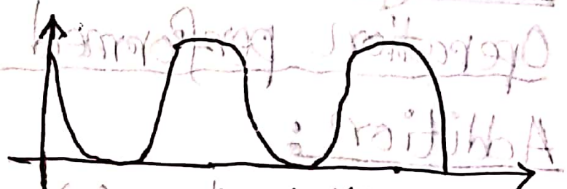
$$y(t) = x_1(t) - x_2(t)$$

For discrete-time signals we write,

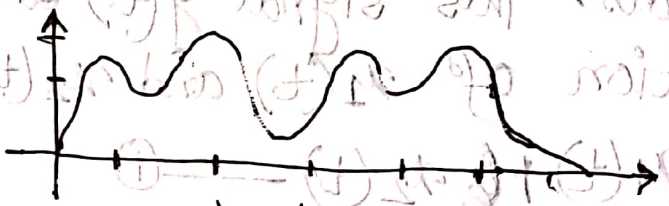
$$y[n] = x_1[n] - x_2[n]$$



signal one



signal two



signal one − signal two (subtraction)

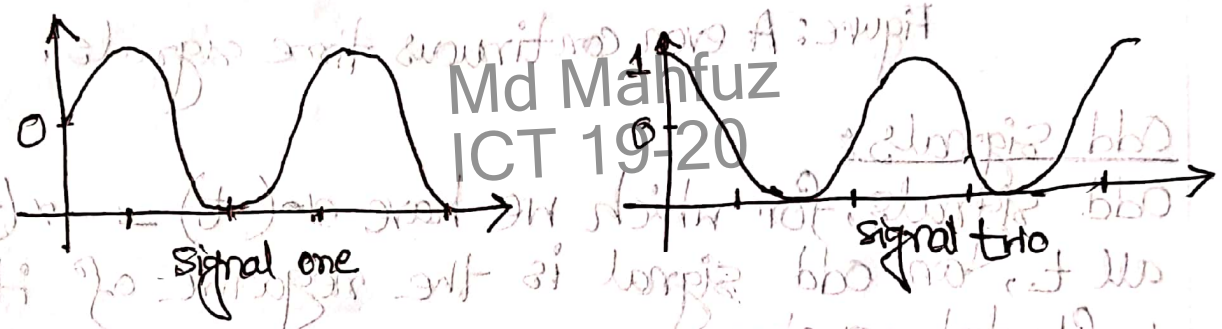
Multiplication: Let $x_1(t)$ and $x_2(t)$ denote a pair of continous time signal. The signal $y(t)$ resulting from the multiplication of $x_1(t)$ by $x_2(t)$ is defined by

$$y(t) = x_1(t) \cdot x_2(t)$$

That is, for each prescribed time t the value of $y(t)$ is given by the product of the corresponding values of $x_1(t)$ and $x_2(t)$.

For discrete time signals, we write,

$$y[n] = x_1[n] \cdot x_2[n]$$



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Amplitude scaling:

Let $x(t)$ denote a continuous time signal. The signal $y(t)$ resulting from amplitude scaling applied to $x(t)$ is defined by

$$y(t) = cx(t)$$

where c is a scaling factor. According to this equation, the value of $y(t)$ is obtained by multiplying the corresponding value of $x(t)$ by the scalar c.

For discrete-time signals we write,

$$y[n] = cx[n]$$

Even signals:

Even signals, for which we have $x(-t) = x(t)$ fo all t, an even signal is the same as it reflected version.

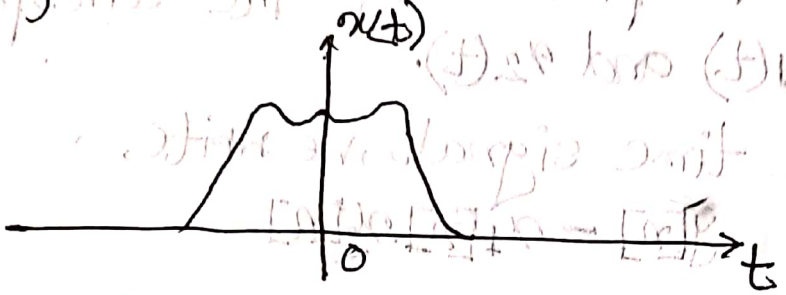

Figure: A even continuous time signals.

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Odd signals:

Odd signals, for which we have $x(-t) = -x(t)$ fo all t, an odd signal is the negative of its reflected version.

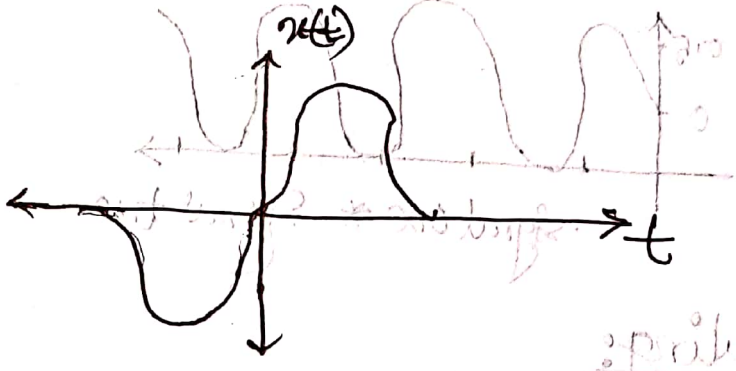

Figure: an odd continuous time signal.

Time scaling:

Let $x(t)$ denote a continous time signal. The signal $y(t)$ obtained by scaling the independen variable, time, t, by a factor a is defined by

$$y(t) = x(at)$$

If $a>1$, the signal $y(t)$ is a compressed version of $x(t)$. If, on the other hand, $0<a<1$, the signal $y(t)$ is an expanded (streched) version of $x(t)$. These two operation illustrated in below:

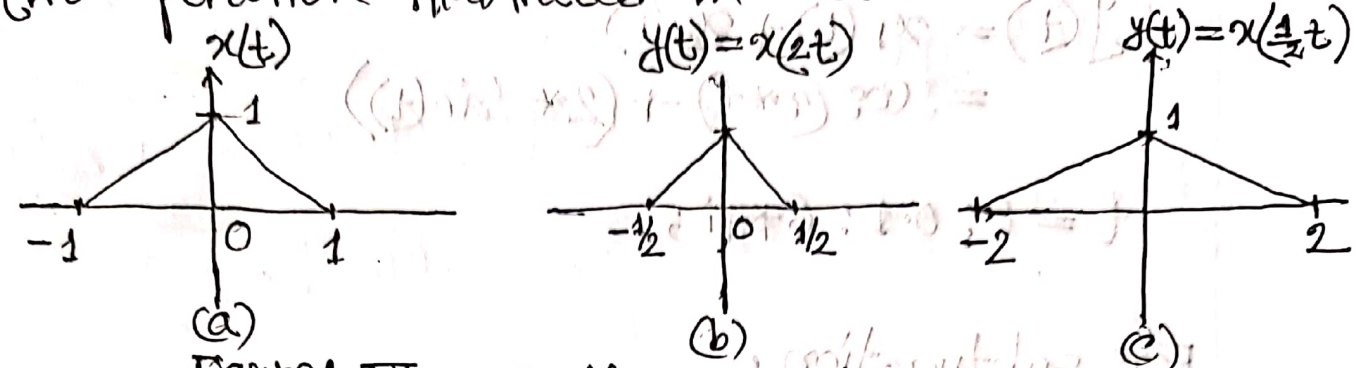

Figure: Time scaling operation

(a) Continous time signal $x(t)$ (b) Compressed version of $x(t)$ by a factor of 2 and (c) expanded version of $x(t)$ by a factor of 2.

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Time Reversal:

Let $x(t)$ denote a continuous-time signal. Let $y(t)$ denote the signal obtained by replacing time t with $-t$, as shown by,

$$y(t) = x(-t)$$

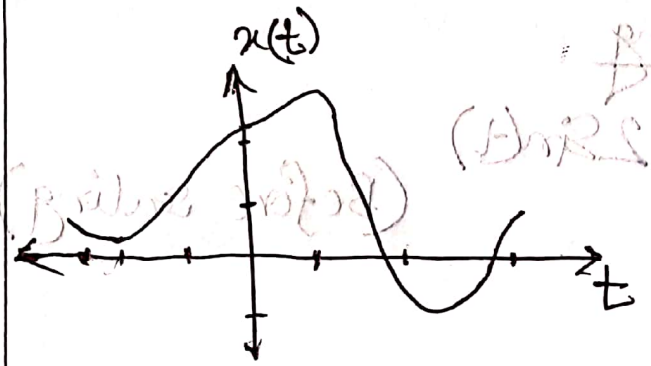

Figure: A Continous time signal $x(t)$

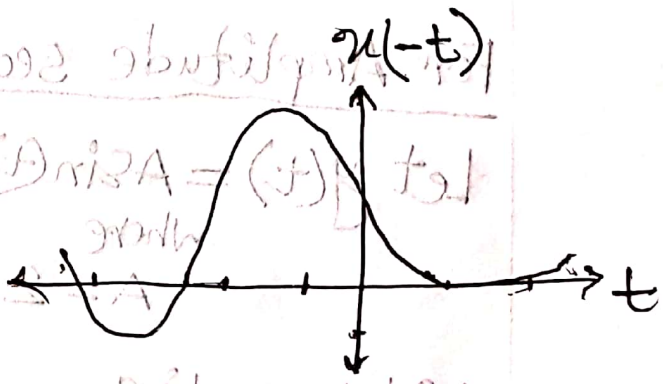

Figure: $x(t)$ reflection version $x(-t)$.

Calculation: (For Addition)

Let two signal $x_1(t) = \cos(3t)$ ——— (i)

$x_2(t) = 2\sin(t)$ ——— (ii)

$\therefore y(t) = x_1(t) + x_2(t)$

$= \cos(3*t) + (2*\sin(t))$

```
t = 0:0.1:6*pi;
```

For subtraction:

$\therefore y(t) = x_1(t) - x_2(t)$

$= \cos(3*t) - (2*\sin(t))$

$x_1(t) = \cos(3t)$

$x_2(t) = 2\sin(t)$

```
t = 0:0.1:6*Pi
```

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For Multiplication

$\therefore y(t) = x_1(t)\, x_2(t)$

$= (\cos(3*t)) * (2*\sin(t))$ [From eq. (i) and (ii)]

```
t = 0:0.1:6*pi
```

For Amplitude scaling

Let $y(t) = A\sin(t) = 2\sin(t)$

where

$A = 2$

(Before scaling)

After scaling,

$y(t) = (1/2) * \sin(t)$ where, $A = 1/2$.

For Even and Odd signals:

```
t = 0:0.1:5
x(t) = t.*exp(t);
xeven = 0.5*t.*(exp(t)-exp(t));
xodd = 0.5*t.*(exp(t)+exp(t));
```

For time scaling

```
t = 0:0.01:20
x = sin(t)
```

Change the independent variables for t/2 and 2t.

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Algorithm:

For Addition, Subtraction, Multiplication,

(i) ~~Steat~~ Start.
(ii) Define the two or more signals.
(iii) Get the value of the range of t.
(iv) Then, calculate the signals as addition/subtraction/multiplication.
(v) Plot the value of the function with t.
(vi) Stop.

For Amplitude Scaling and time scaling,

(i) Start
(ii) Define the ~~value of~~ function with amplitude and t.
(iii) Peak a value of amplitude with desired amplitude
(iv) Plot the value of the function with t.
(v) Stop.

For even and odd signal;

(i) Start
(ii) Define a function of $x(t)$.
(iii) Get the value of range t.
(iv) Define, $\frac{1}{2}t.(\exp(t)-\exp(t))$ (For even signal);
and
$\frac{1}{2}t.(\exp(t)+\exp(t))$ (For odd signal)
(v) Plot the value of $x_{even/odd}$ with t.
(vi) Stop.

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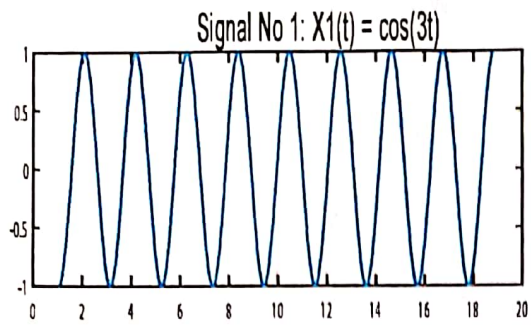
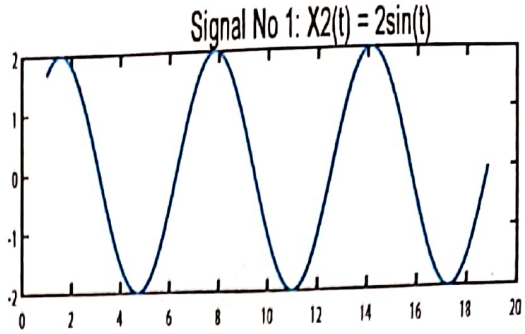
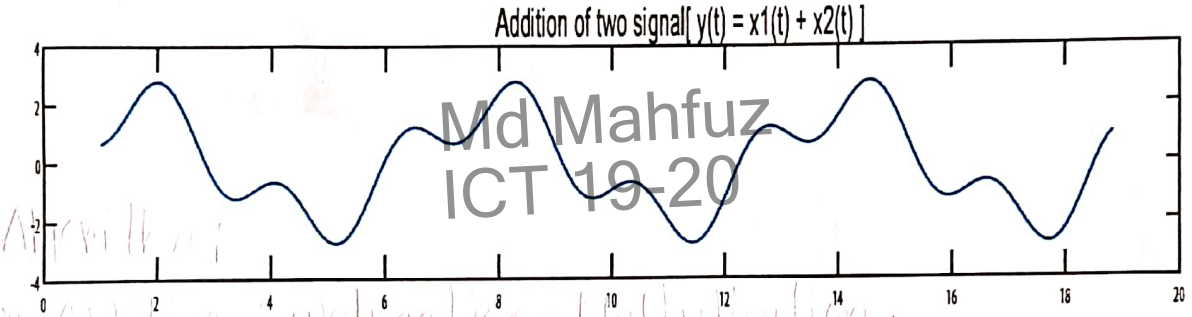
Output







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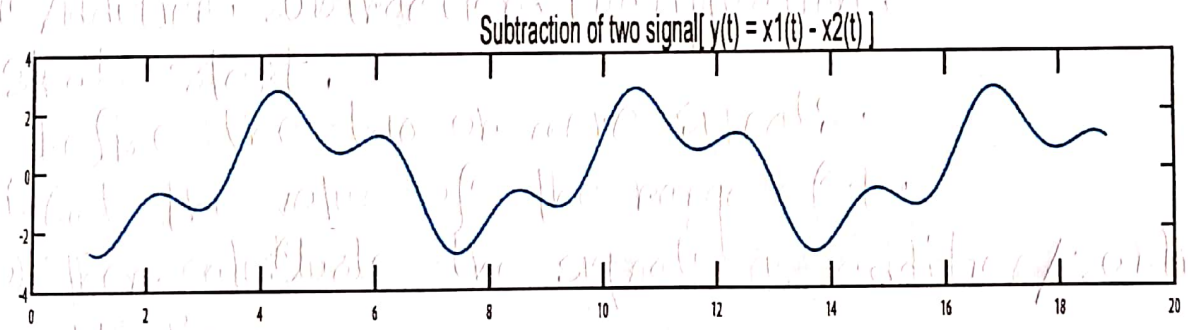

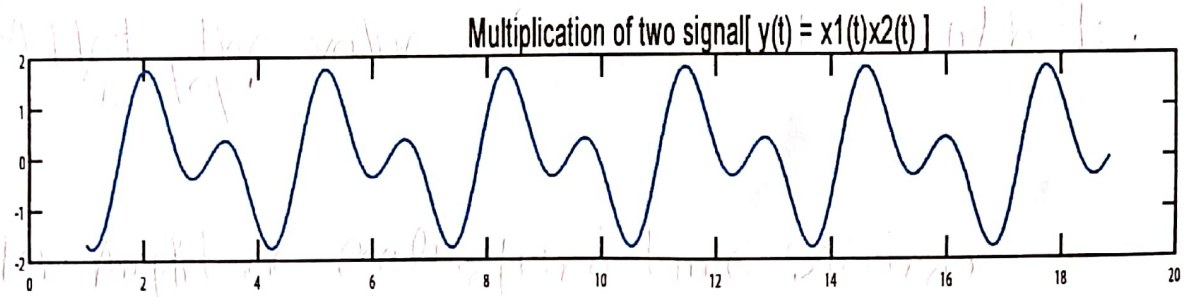

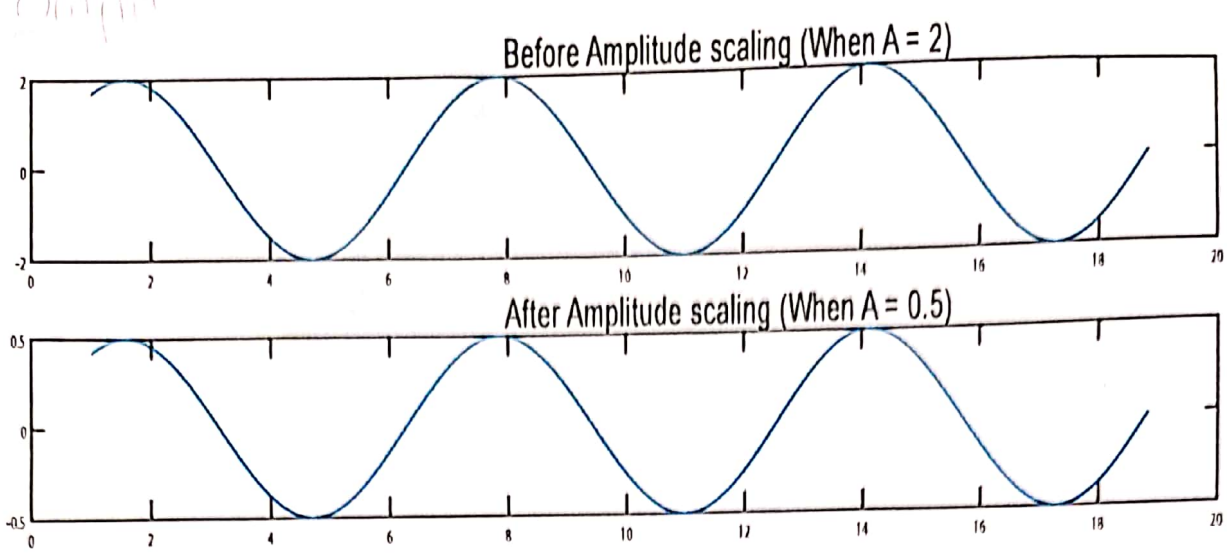

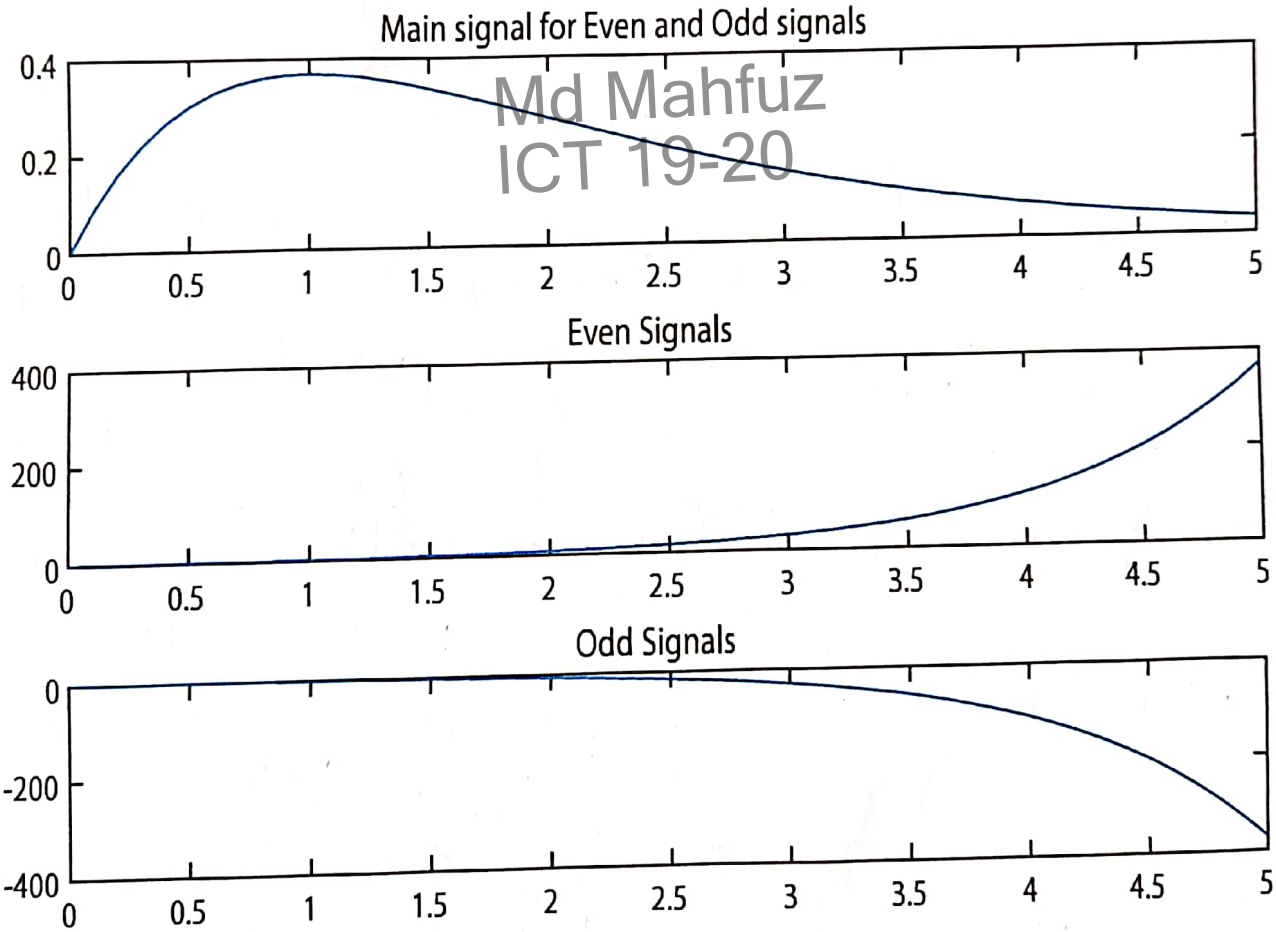

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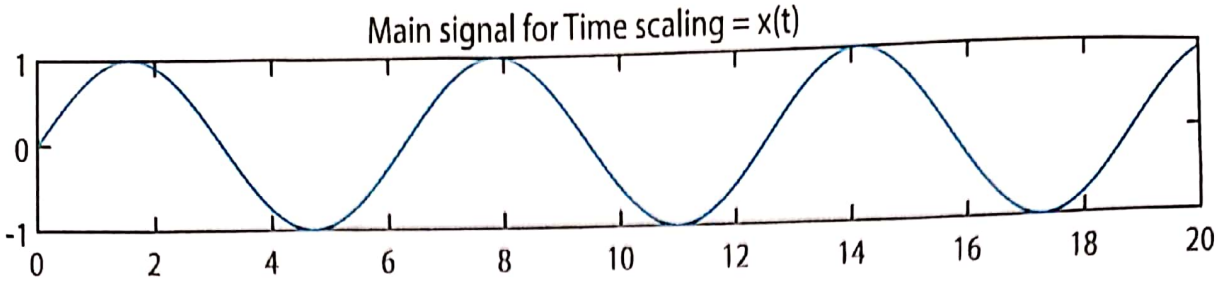

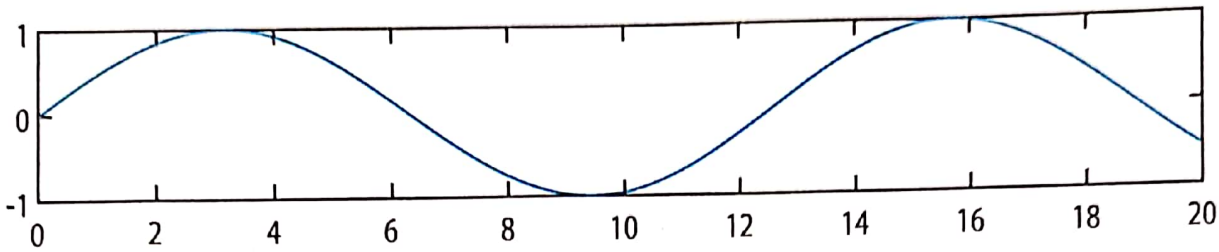

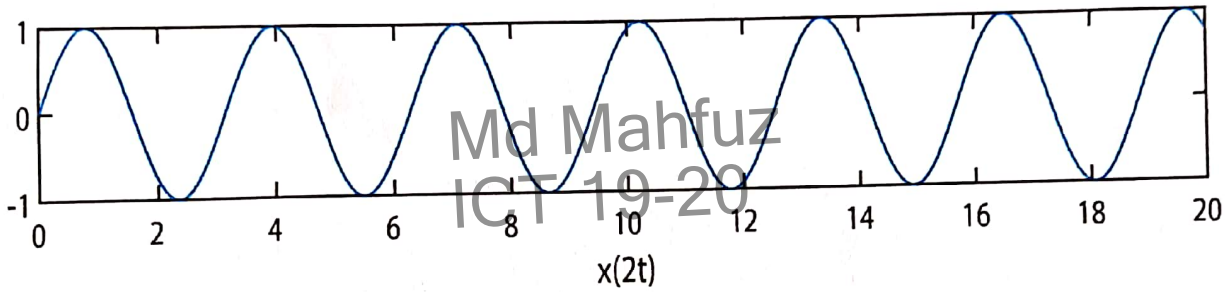

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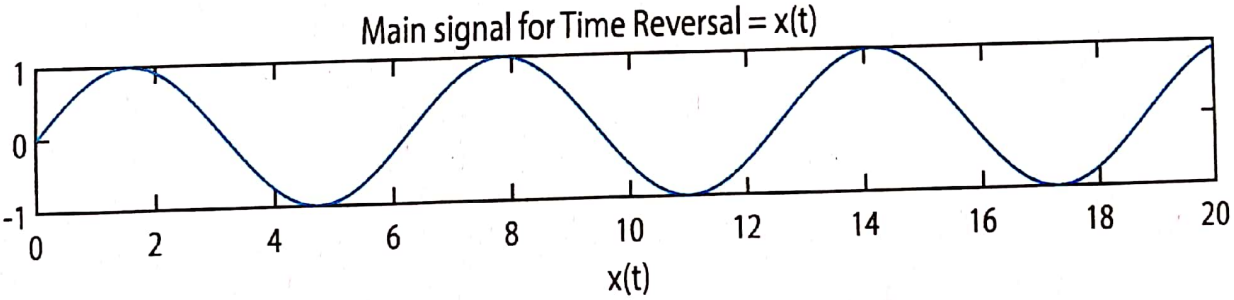

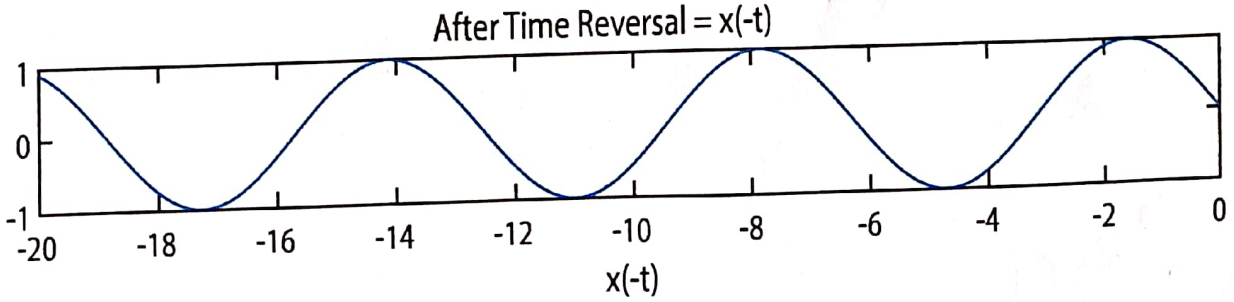

Result and Disscussion:

(i) Addition, subtraction, multiplication two or more signal for useful devices like as Audio Mixer, radio signal etc. to implement this signal.

(ii) To decrease or increase playback speed of a signal by using the time scaling factor.

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Experiment No: 3

Name of the experiment: Write a program to verify the following properties of a given system in continuous/discrete time (i) Causal and Non-causal systems. (ii) Static (Memory less) and Dynamic (with memory (iii) Linear and Non-Linear systems and (iv) Time-Invariant and Time-Varient system.

AIM:

To represent the properties of a given systems.

Objectives

After completing this experiment, we are able to know about this:

(i) How to working with the past, present and future inputs of signals.

(ii) know about the different kinds of signals.

(iii) Implement the course concepts on using MATLAB.

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Theory:

Causal and Non-Causal systems:

A system is said to be causal if its output depends upon present and past inputs, and does not depend upon future input.

For non-causal system, the output depends upon future input also.

Example 1: $y(n) = 2x(t) + 3x(t-3)$

For present value $t=1$, the system output is —

$y(1) = 2x(1) + 3x(-2)$.

Here, the system output only depends upon present and past inputs. Hence, the system is causal.

Example 2: $y(n) = 2x(t) + 3x(t-3) + 6x(t+3)$

~~Here, the system~~ For present value $t=1$, the system output is $y(1) = 2x(1) + 3x(-2) + 6x(4)$. Here, the system output depends upon future input. Hence the system is non-causal system.

Static (Memoryless) and Dynamic (with Memory) systems:

static system is memoryless whereas dynamic system is a memory system.

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Example 1: $y(t) = 2x(t)$.

For present value $t=0$, the system output $y(0) = 2x($

Here, the output only dependent upon present input

Hence the system is memoryless or static.

Example 2: $y(t) = 2x(t) + 3x(t-3)$

For present value $t=0$, the system output is

$$y(0) = 2x(0) + 3x(-3)$$

Here, $x(-3)$ is past value for present input for which the system requires memory to get this output. Hence, the system is a dynami system.

Linear and Non-linear systems

A system is said to be linear when it satisfies superposition and homogenate principles. Consider two systems with inputs as $x_1(t)$, and outputs as $y_1(t)$, $y_2(t)$ respectively. Then, according to the superposition and homogenate principles,

$$T[a_1x_1(t) + a_2x_2(t)] = a_1T[x_1(t)] + a_2T[x_2(t)]$$

$$\therefore T[a_1x_1(t) + a_2x_2(t)] = a_1y_1(t) + a_2y_2(t)$$

From the above expression, is clear that response of overall system is equal to response of individual system.

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Example

$y(t) = x^2(t)$

Solution:

$$y_1(t) = T[x_1(t)] = x_1^2(t)$$

$$y_2(t) = T[x_2(t)] = x_2^2(t)$$

$$T[a_1x_1(t) + a_2x_2(t)] = [a_1x_1(t) + a_2x_2(t)]^2$$

Which is not equal to $a_1y_1(t) + a_2y_2(t)$. Hence, the system is said to be nonlinear.

Time Varient and Time Invarient systems:

A system is said to be time varient if its input and output characteristics vary with time. Otherwise, the system is considered as time invariant. The condition for time invarient system is:

$$y(n,t) = y(n-t)$$

The condition for time varient system is

$$y(n,t) \neq y(n-t)$$

Where, $y(n,t) = T[x(n-t)]$ = input change.

$y(n-t)$ = output change.

Example:

$y(n) = x(-n)$

$y(n,t) = T[x(n-t)] = x(-n-t)$

$y(n-t) = x(-(n-t)) = x(-n+t)$

$\therefore y(n,t) \neq y(n-t)$. Hence the system is time varient.

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Calculation:

Let the causual system equation ——
$y(t) = x(t)$ when $t=0$;

(For causual an
Non-causal sy

Note that, the output depends on the present input

Another,

$y(t) = x(t-1)$

When, $t=-1$, the outputput depends on the both present and past inputs.

Let the Non-causual system equation ——
$y(t) = x(t+1)$ which output depend on ~~pre~~ future input.

Another,

$y(t) = x(t) + x(t-1) + x(t+1)$ which the output present, past and future inputs.

$t = [-3, 3]$ and $n = -3:0.5:3$

For static and Dynamic system:

Let a equation, $y(t) = x(t)$ when the input value doesn't be changeable ——

For Dynamic or with memory system,

Let a equation, $y(t) = k \cdot x(t)$ where $k = 0, 1, 2 \cdots n$

$t = -3:0.1:3$, and $n = -3:0.5:3$

For Time-Invariant and Time Variant systems:

Consider a system, described by the i/o relationship $y(t) = 1 - 2x(t-1)$;

Determine if this is a time-invariant system by using the input signal

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$x(t) = \cos(t)\,[u(t) - u(t-10)]$;

$y(t) = 1 - 2x(t-1) * x(t)$;

$t = -5:0.01:20$; and $n = -5:0.5:20$

For Linear and Non-Linear Systems:

For linear system ——

$$S\{a_1 x_1(t) + a_2 x_2(t)\} = a_1 S\{x_1(t)\} + a_2 S\{x_2(t)\}.$$

Let $x_1(t) = u(t) - u(t-1)$ and $x_2(t) = u(t) - u(t-2)$ be input signals to the systems described by the i/o relationships $y(t) = 2 * x(t)$ and $y(t) = x2(t)$.

To examine if the system are Linear, we use the scalars $a_1 = 2$ and $a_2 = 3$.

$t = -3 \leq t \leq 3$.

$n = -3:0.5:3$

Algorithm:

Algorithm for causal and non-causal systems,

(i) Start

(ii) Taken the signals which zero and one matrix — $x_1(t)$ and $x_2(t)$ or more ....

(iii) Define the range of value t for every matrix.

(iv) Combine the signals and time to signal and ti

(v) If $t<=0$

This is a causal system

else

This is Non-causal syston

(vi) Plotting the signal respect to the time

(vii) Stop.

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For Static and Dynamic Systems,

(i) Start

(ii) Given the signal which zero and one matrix — $x_1(t)$ and $x_2(t)$ or more ——

(iii) Define the range of t for every matrix.

(iv) Combine the signals and time to a signal and

(v) If $t=0$ or fixed

This is a static system

else $(t<0)$

This is a dynamic system.

(vi) Plotting the signal respect to the time

(vii) Stop.

For Linear and Non-linear system;;

(i) Start

(ii) Define the linear system —

$$S\{a_1 x_1(t) + a_2 x_2(t)\} = a_1 S\{x_1(t)\} + a_2 S\{x_2(t)\}$$

(iii) Define the input signals for unit step function

(iv) Let a signal as input/output relationships.

(v) Plotting the signal respect to the time.

(vi) Stop.

For Time-Invariant and Time Variant systems,

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(i) Start

(ii) Define a signal

(iii) Define the input signal as unit step function

(iv) Multiply the two signals.

(v) If (time does not change respect to time)

It is Time Invarient signals

else

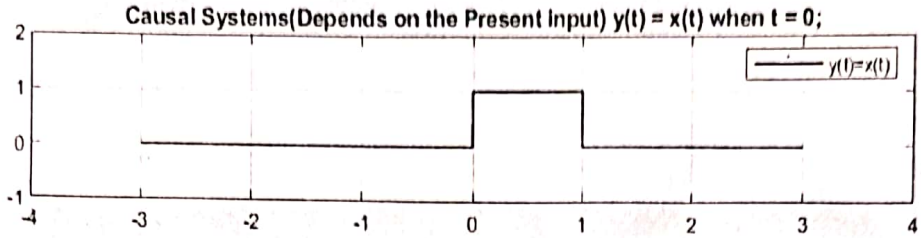
It is Time Invarient signals

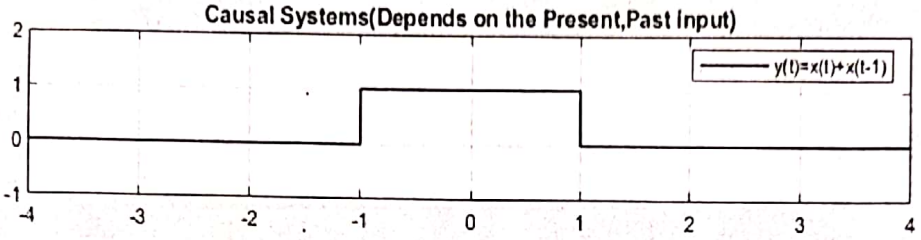
(vi) Plotting the signal respect to the time

(vii) Stop.

Output:

For Continuous time Signals:





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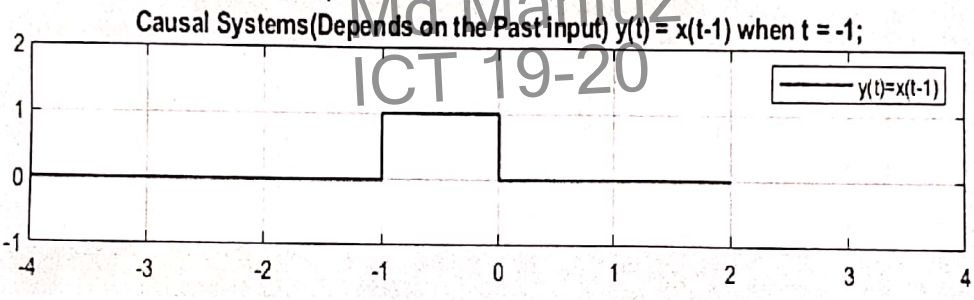
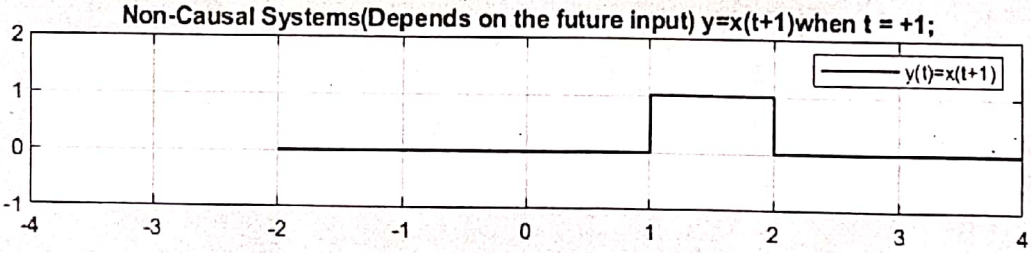
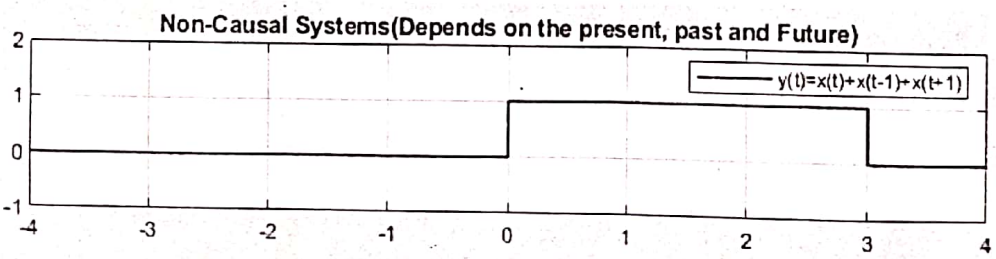
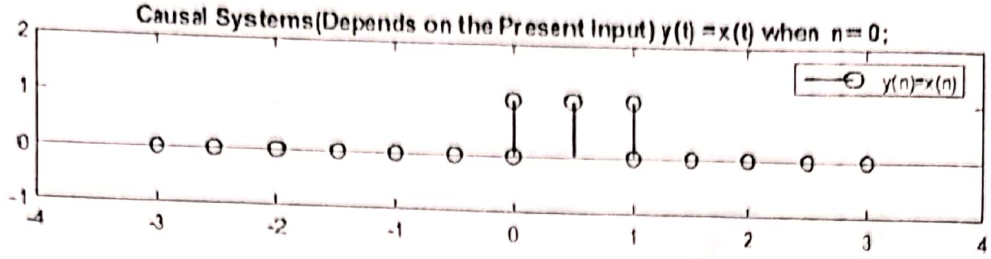
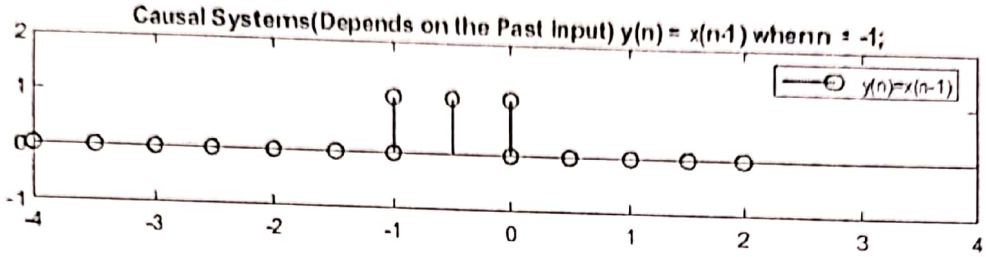


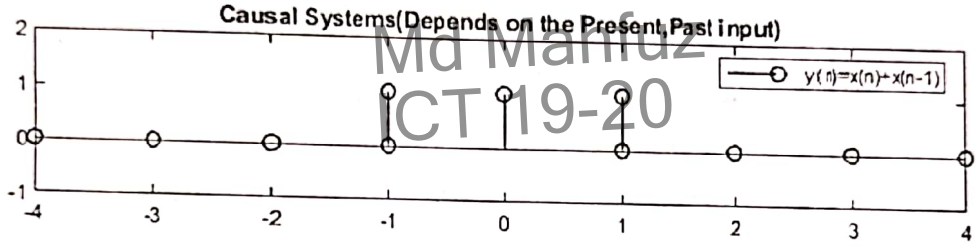




For Discrete Time Signals:







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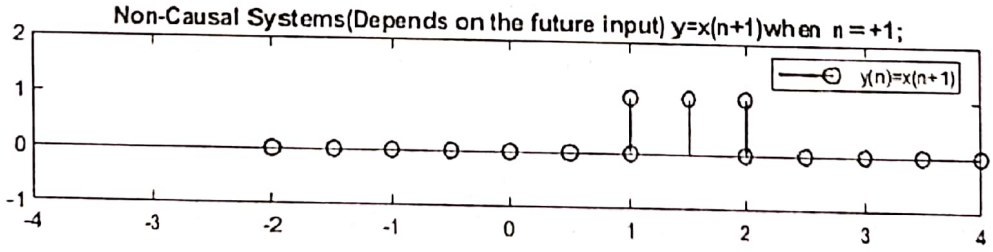
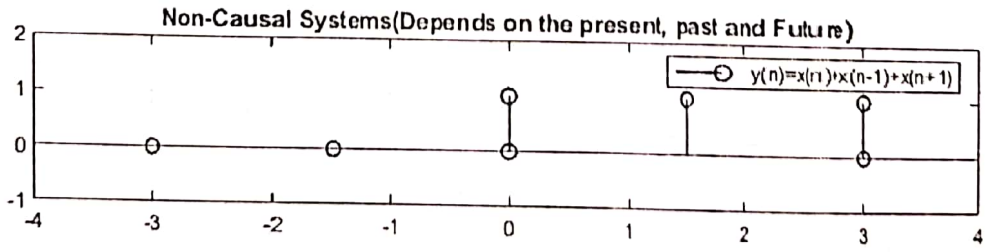
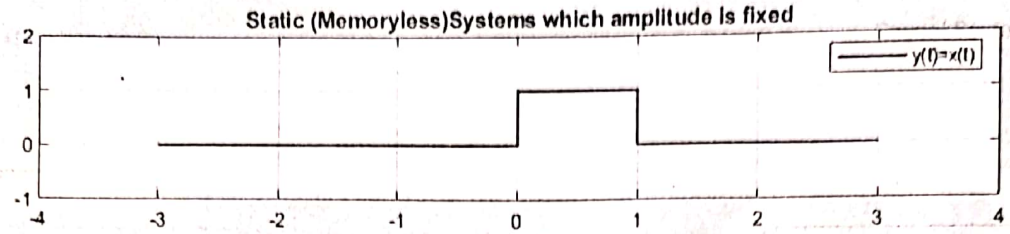
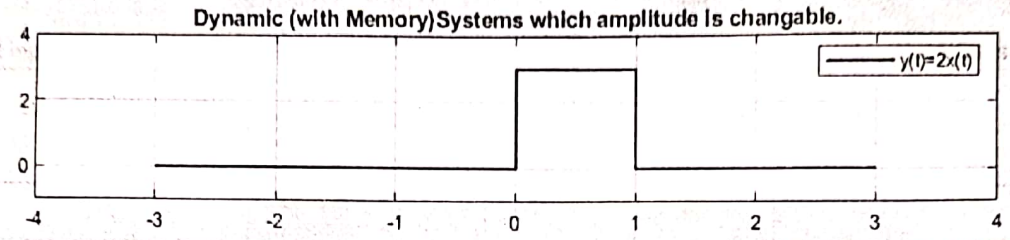




For Continuous Time Signals,





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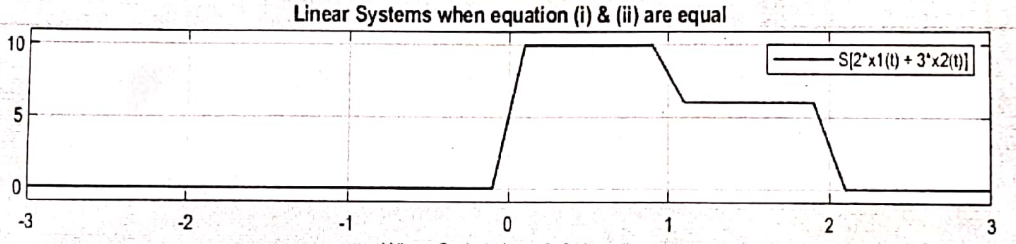
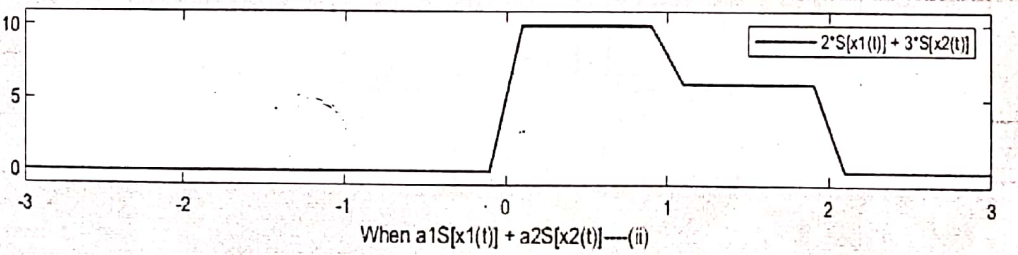
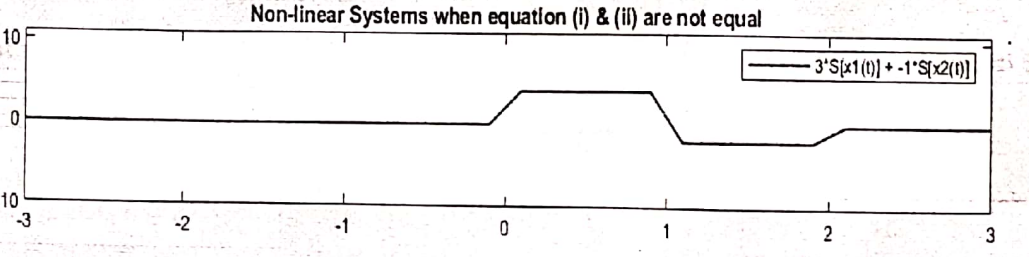


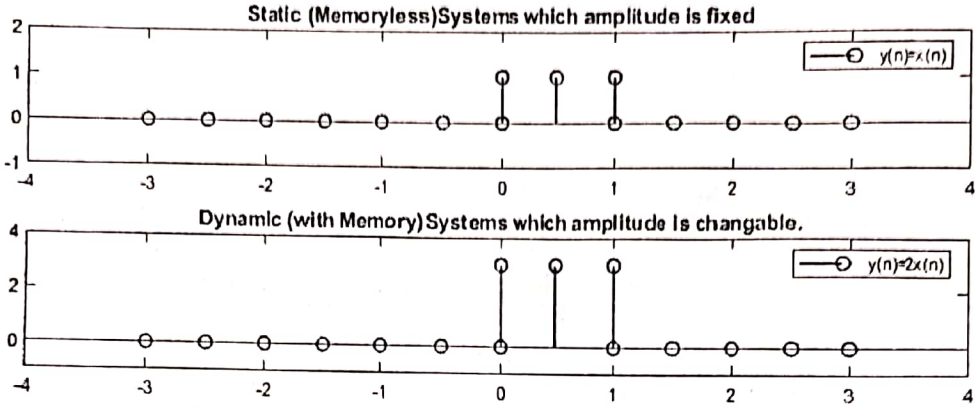
When Sa1x1(t) + a2x2(t)----(i)



When a1S[x1(t)] + a2S[x2(t)]----(ii)



For Discrete Time Signals:



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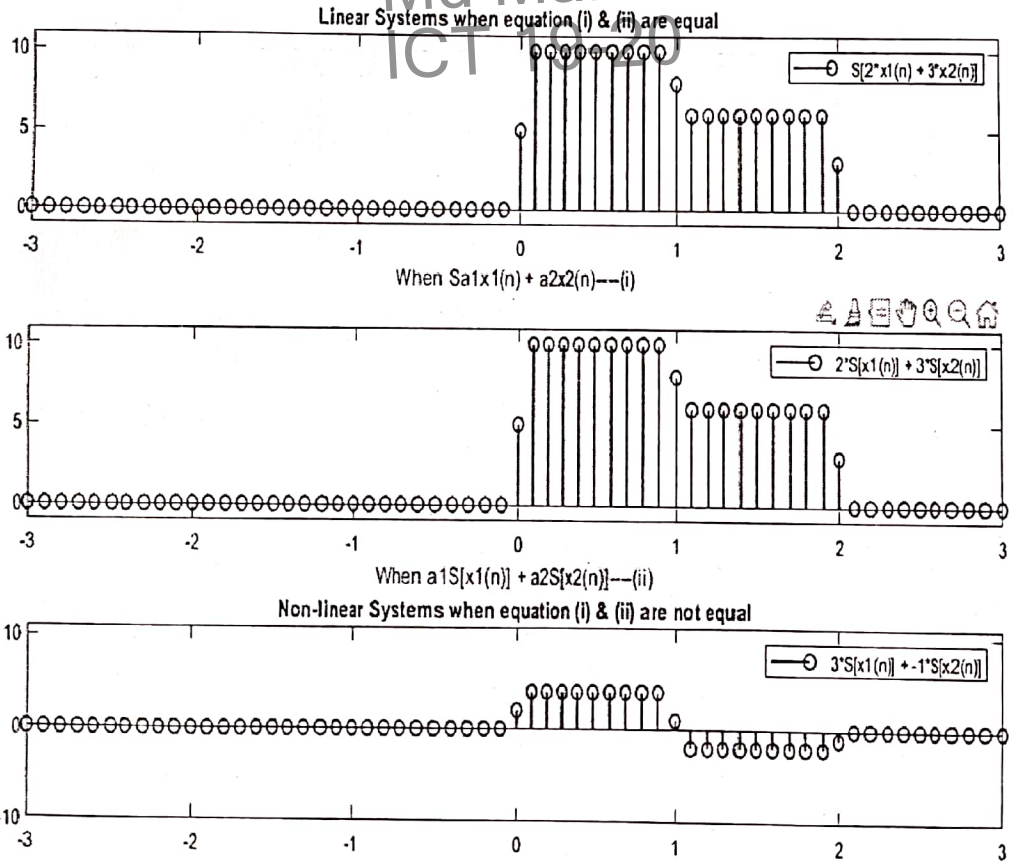
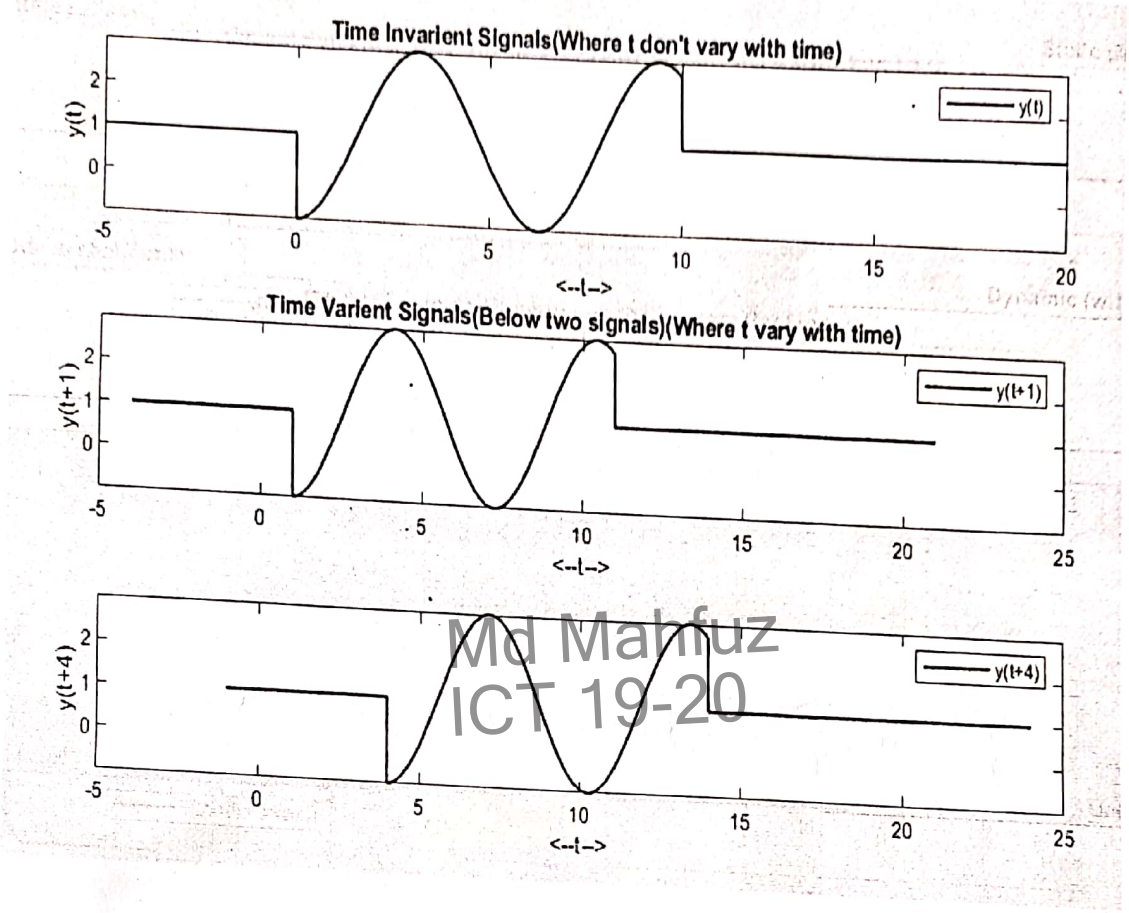


For Continuous Time Signals:



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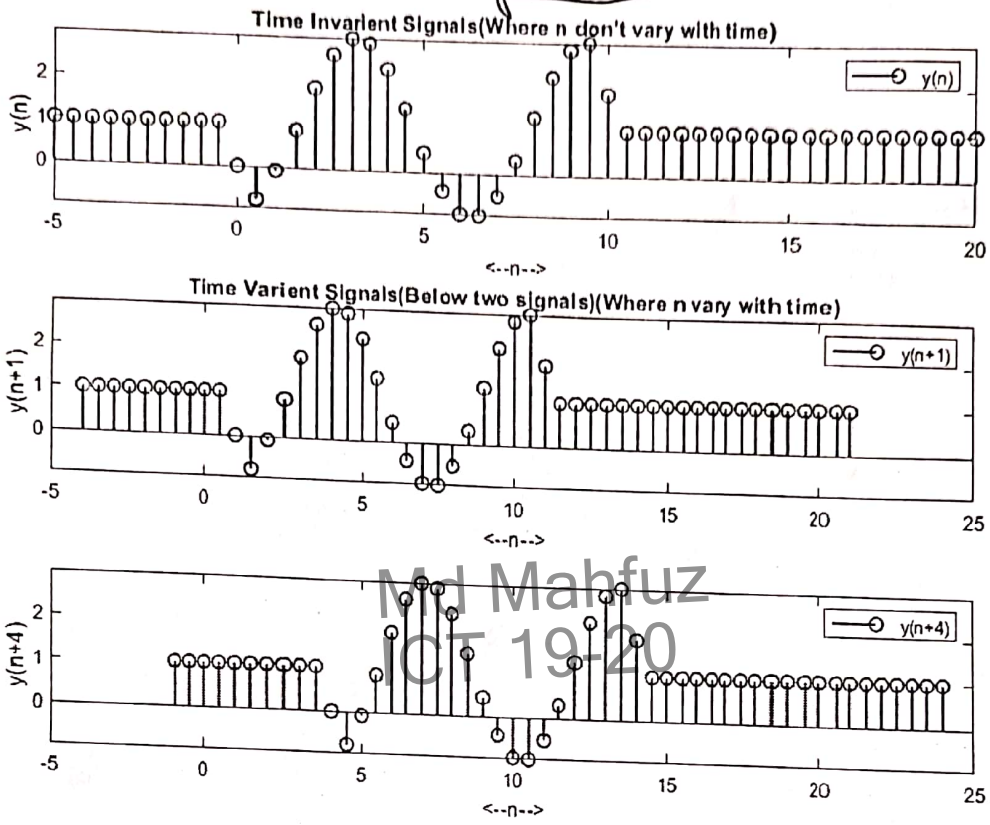
For discrete time Signals:



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Result and Discussion:

(i) If the value of $t<=0$ that depends on present and past input, it is called a causal system and $t>0$ or depends on future value, it is called Non-causal system.

(ii) If the value of $t=0$ that depends on only present input which called static or Memoryless system and depends on past input for with memory or dynamic system.

(iii) A system be linear what it satisfied super-position and homogenate principles either non-linear.

(iv) If a system input and output vary with time, this is called time variant system either Time Invarient system.

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Experiment No: 4 & 5

Name of the experiment: Write a program to convolve two continuous/Discrete time signals. Plot all the signals. Verify the result by analytical calculation.

AIM:

To perform convolution of continuous time signals.

Objectives:

After completing this experiment, we are able to know about this:

(i) Know about impulse response and convolution.
(ii) Implement the course concepts on using MATLAB.

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Theory:

Convolution: Convolution is a mathematical tool to combining two signals to form a third signal. Therefore, in signals and system, the convolution is very important because it relates the input signal and the impulse response of the system to produce the output signal from the system. In other words, the convolution is used to express the input and output relationship of an LTI system.

Explanation: Consider a continuous-time LTI system which is relaxed at $t=0$, i.e., initially, no input is applied to it. Now, if the impulse signal $\delta(t)$

is input to the system, then output of the system is called the impulse response $h(t)$ of the system and is given by,

$$h(t) = T[\delta(t)]$$

As any arbitray signal $x(t)$ can be represented

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,\partial\tau$$

Then the output of the system corresponding to $x(t)$ is given by,

$$y(t) = T[x(t)]$$

$$\Rightarrow y(t) = T\left[\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,\partial\tau\right]$$

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For a continuous-time linear system, the outp is given by,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\,T[\delta(t-\tau)]\,d\tau \quad \text{——(i)}$$

Now, if the response of the system due to impulse signal $\delta(t)$ is $h(t)$, then the respor of the linear system due to delayed impulse signal is given by,

$$h(t,\tau) = T[\delta(t-\tau)]$$

By substituting the value of $T[\delta(t-\tau)]$ in the equation (i), we get,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t,\tau)\,d\tau \quad \text{——(ii)}$$

Again, for a time-invariant system, the output corresponding to the input delayed by T units is equal to the output delayed by T units, i.e.,

$$h(t,\tau) = h(t-\tau)$$

On putting the value of $h(t,\tau)$ in the equation (ii), we have,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau \quad \text{——— (iii)}$$

The expression in equation (iii) is called the convolution integral or simply convolution.

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Therefore, the convolution of two continuous time signals $x(t)$ and $h(t)$ is represented as,

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

Convolution is a mathematical operation used to express the relation between input and output of an LTI system. It relates input, output and impulse response of an LTI system.

$$y(t) = x(t) * h(t)$$

where,

$y(t)$ = output of LTI

$x(t)$ = input of LTI

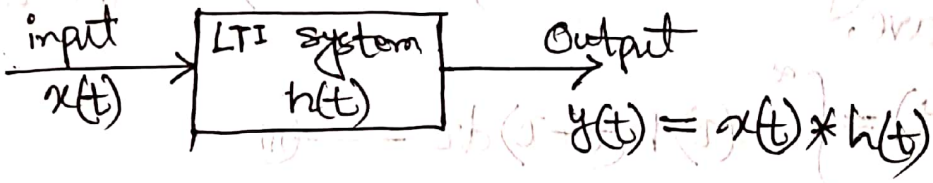
$h(t)$ = impulse response of LTI

There are two types of convolutions:

(i) Continuous convolution

(ii) Discrete convolution

Continuous Time Convolution:



$$y(t) = x(t) * h(t)$$

$$= \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

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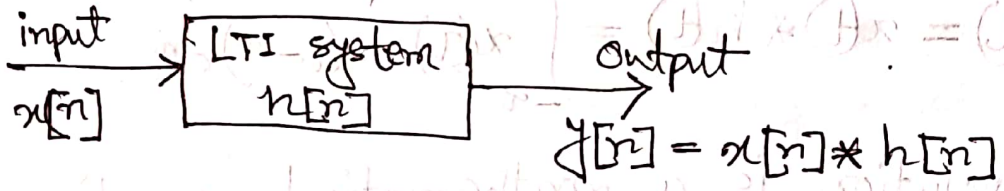
Discrete Time Convolution:



$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[n-k]\, h[k].$$

By using convolution, we can find zero state response of the system.

CALCULATION: Example 2.6

Input signal → $x(t) = e^{-at} \cdot u(t)$ when $a>0$ and

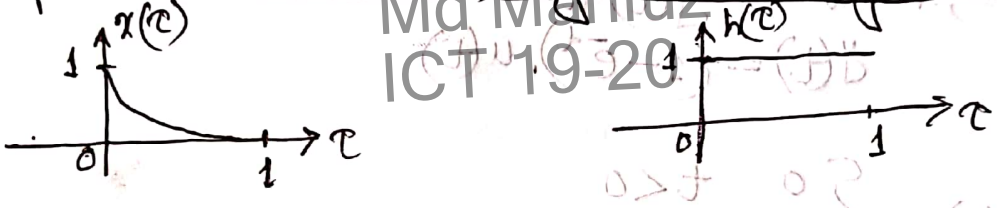
Impulse Response → $u(t)$

Now, putting the value of $a=1$ which $a>0$ and we get the signal;

$$x(t) = e^{-t} \cdot u(t) \qquad 0<t<1$$
$$h(t) = u(t) \qquad -1<t<1$$

Now, the way of finding convolution for this signals will be started in this below:

① Replace t with τ (tau) by those signals:



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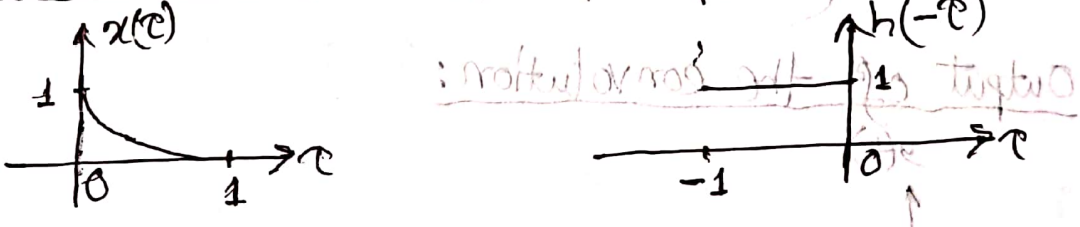
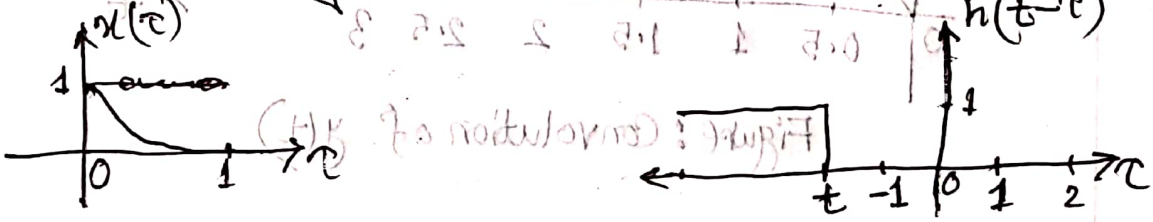
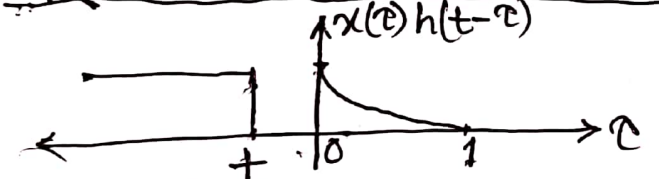
② Time reversal for impulse response $h(0-\tau)$:



③ Time shifting for impulse response $h(t-\tau)$:



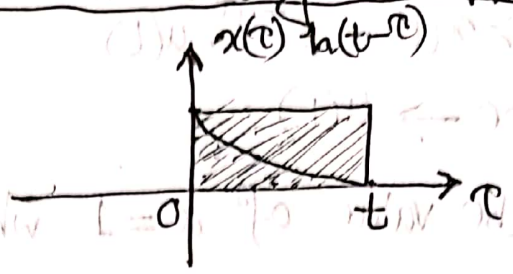
④ Case 1: $t<0$, and convolution starting $-\infty$ to $+\infty$



Here, there are no overlapping.

$\therefore y(t) = 0 \int_{-\infty}^{0} e^{-\tau} d\tau = [-e^{-\tau}]_{-\infty}^{0} =$ Not possible case $= 0$.

(ii) Case 2: $t>0$ and starting overlapping:



$$\therefore y(t) = \int_0^t x(\tau)\, h(t-\tau)\, d\tau$$

$$= \int_0^t e^{-\tau} \cdot 1 \, d\tau$$

$$= \left[-e^{-\tau}\right]_0^t = -e^{-t} + e^0 = 1 - e^{-t}$$

Thus, for all t, $y(t)$ is

$$y(t) = (1-e^{-t})\, u(t)$$

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$$\therefore y(t) = \begin{cases} 0 & t<0 \\ 1-e^{-t} & t>0 \end{cases}$$

Output of the convolution:

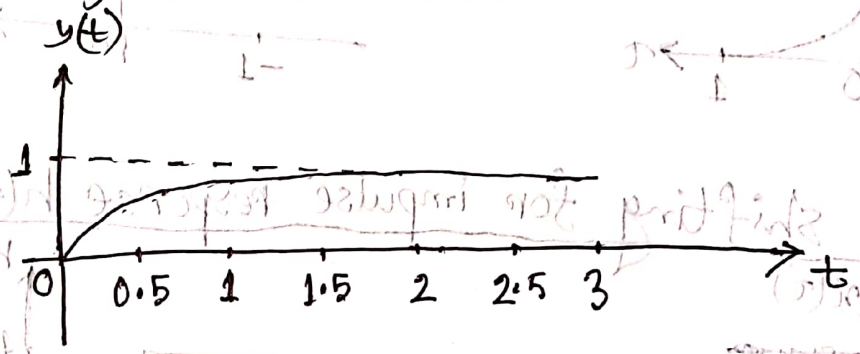
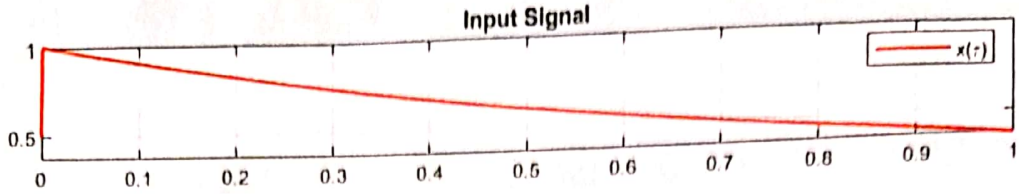
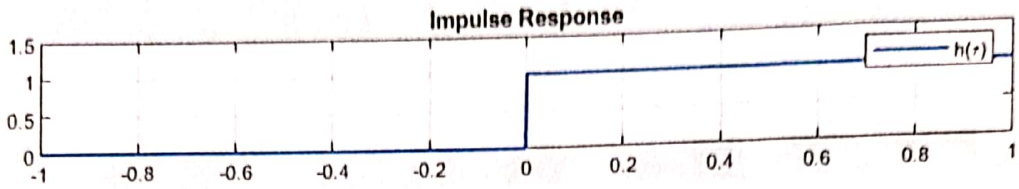
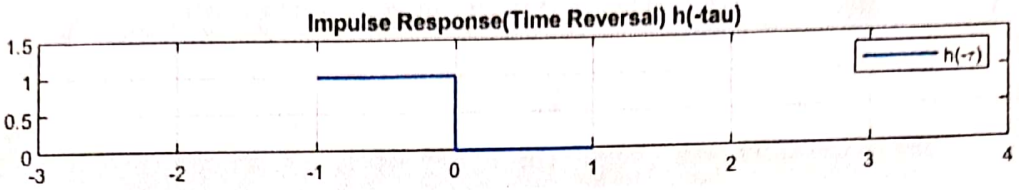


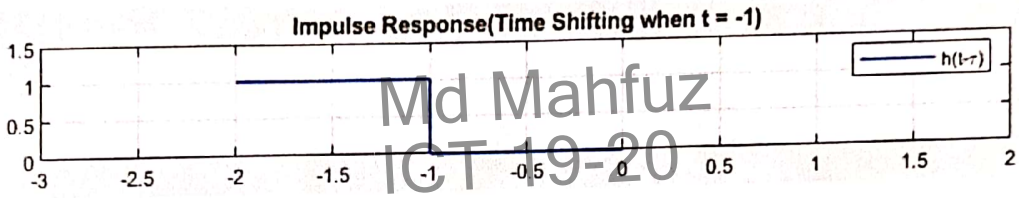
Figure: Convolution of $y(t)$

Output :









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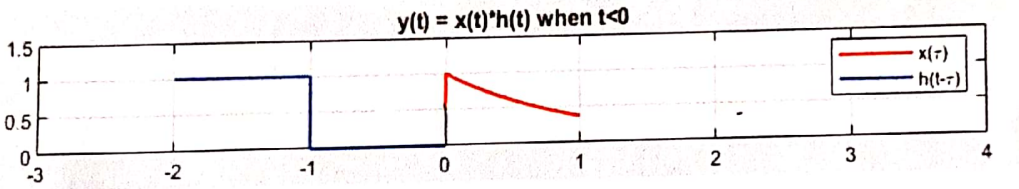

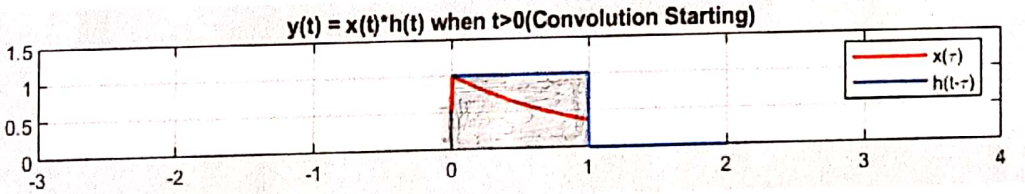

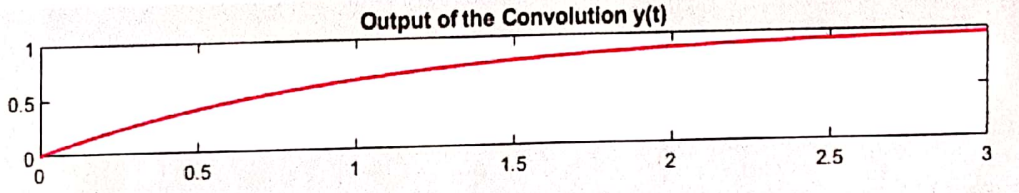

Example 2.7

Calculation:

Consider the convolution of following two signals:

$$x(t) = \begin{cases} 1 & 0 < t < T \\ 0 & \text{otherwise} \end{cases}$$

$$h(t) = \begin{cases} t & 0 < t < 2T \\ 0 & \text{otherwise} \end{cases}$$

Now, putting the value of $T = 1$ and we get the signals

Input signal: $x(t) = 1 \quad 0 < t < 1$

Impulse Response: $h(t) = t \quad 0 < t < 2$

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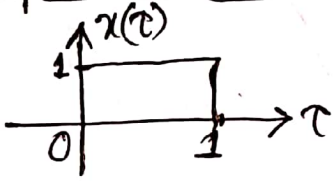
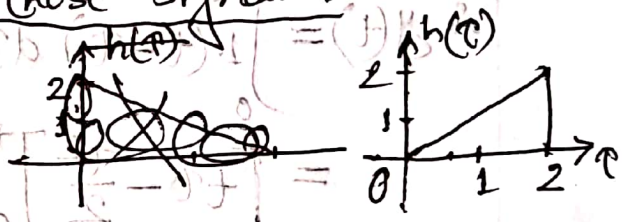
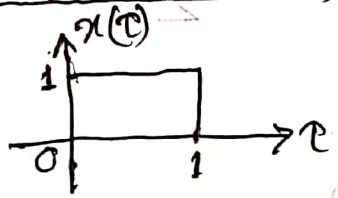
Now, the way of finding convolution for this signals will be started in this below:

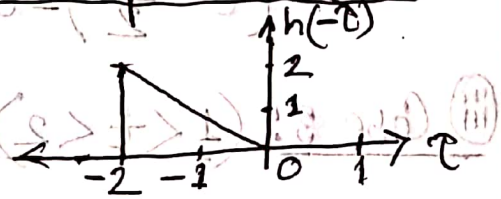
① Replace t with (τ) for those signals:



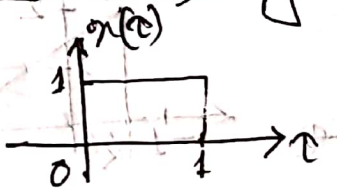


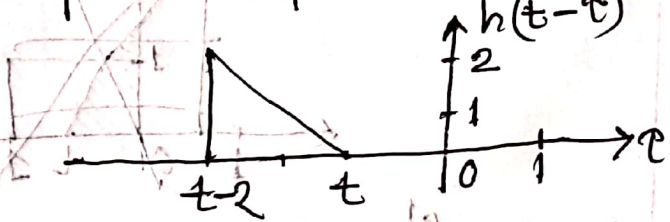
② Time Reversal for impulse response $h(-\tau)$:



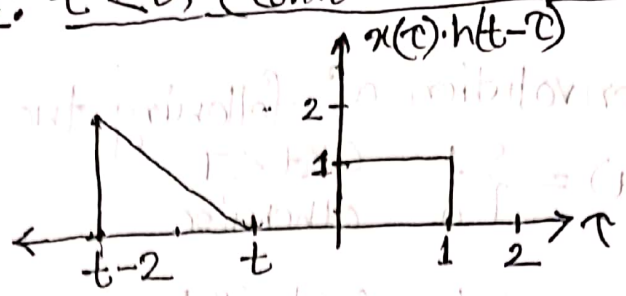


③ Time shifting for impulse response $h(t-\tau)$:





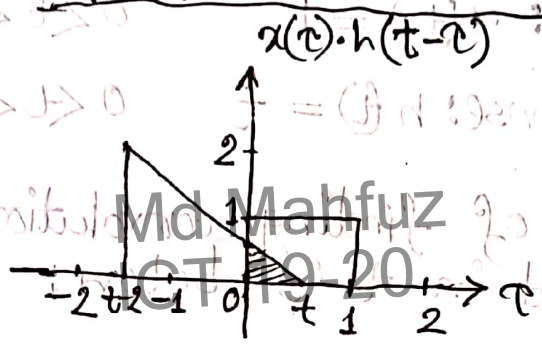
④ (i) Case 1: $t < 0$, (Convolution starting from $-\infty$ to $+\infty$)



Here, there are no overlapping.

$\therefore y(t) = 0.$

(ii) Case 2: $0 < t < 1$ (partial overlapping)



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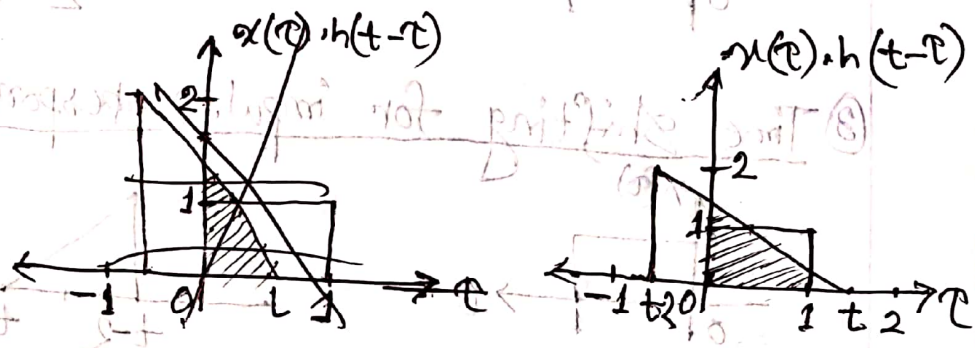
$$\therefore y(t) = \int_0^t 1 \cdot (t-\tau)\, d\tau$$

$$= \left[t\tau - \frac{\tau^2}{2}\right]_0^t$$

$$= \left(t^2 - \frac{t^2}{2}\right) - 0 = \frac{2t^2 - t^2}{2} = \frac{t^2}{2}$$

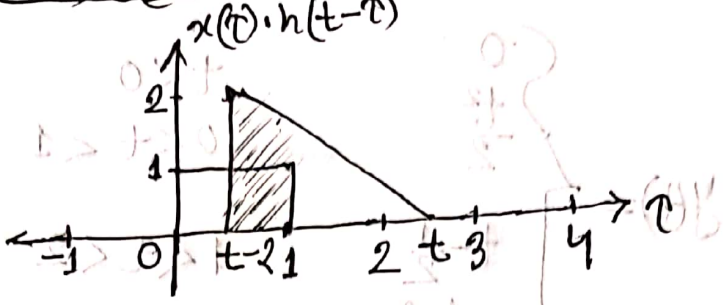
(iii) Case 3: $(1 < t < 2)$



$$\therefore y(t) = \int_0^1 (t-\tau)\, d\tau$$

$$= \left[t\tau - \frac{\tau^2}{2}\right]_0^1 = \left[t \cdot 1 - \frac{1^2}{2}\right] - 0 = t - \frac{1}{2}.$$

(iii) Case 4: $(2<t<3)$ (Exit - partial overlapping)



$$\therefore y(t) = \int_{t-2}^{1} (t-\tau)\, d\tau$$

$$= \left[t\tau - \frac{\tau^2}{2}\right]_{t-2}^{1}$$

$$= \left[t.1 - \frac{1^2}{2}\right] - \left[t(t-2) - \frac{(t-2)^2}{2}\right]$$

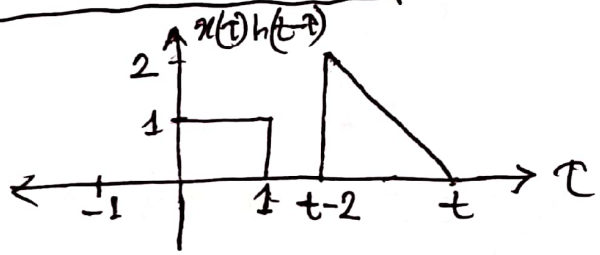
$$= \left[t - \frac{1}{2}\right] - \left[t^2 - 2t - \frac{(t^2-4t+4)}{2}\right]$$

$$= \left[\frac{2t-1}{2}\right] - \left[\frac{2t^2-4t-t^2+4t-4}{2}\right]$$

$$= \frac{2t-1}{2} - \frac{t^2-4}{2}$$

$$= \frac{2t-1-t^2+4}{2} = \frac{2t-t^2+3}{2} = -\frac{1}{2}t^2 + t + \frac{3}{2}$$

(v) Case 5: $3<t$ and zero overlap



Here, there are no overlapping.

$\therefore y(t) = 0.$

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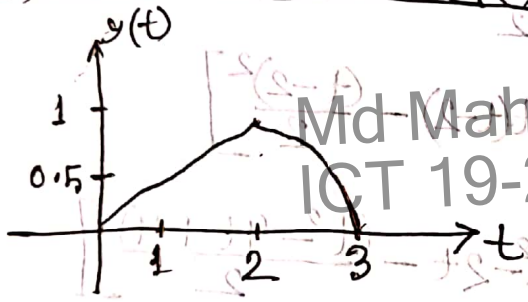
Now, final output after convolution:

$$y(t)=\begin{cases}0 & t<0\\ \frac{t^2}{2} & 0<t<1\\ t-\frac{1}{2} & 1<t<2\\ -\frac{1}{2}t^2+t+\frac{3}{2} & 2<t<3\\ 0 & 3<t\end{cases}$$

Output of the Convolution $y(t)$:



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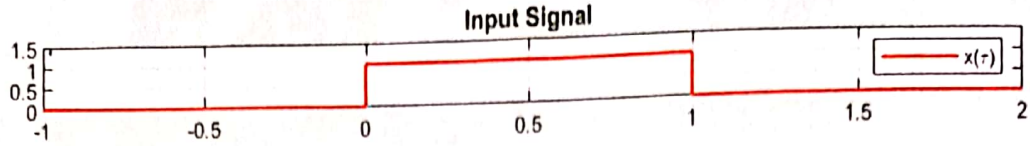
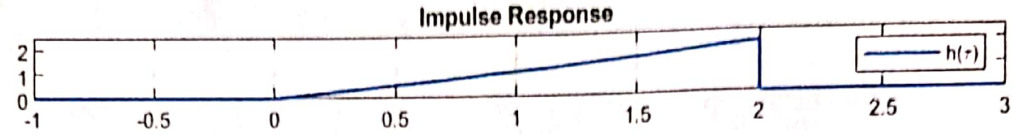
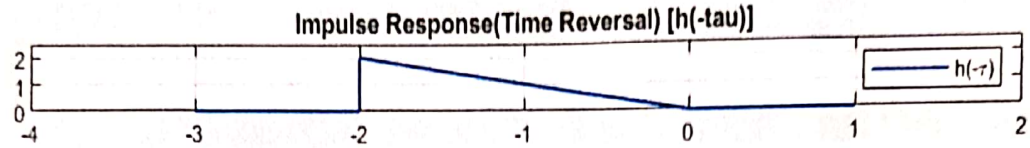
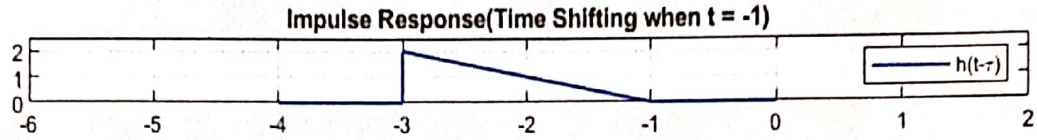
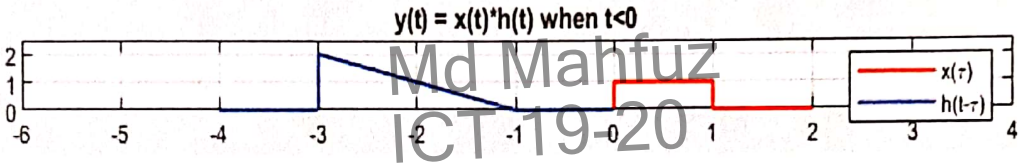
Output:











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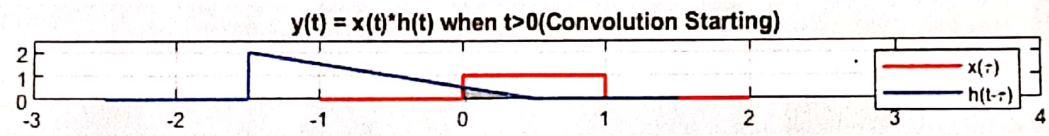
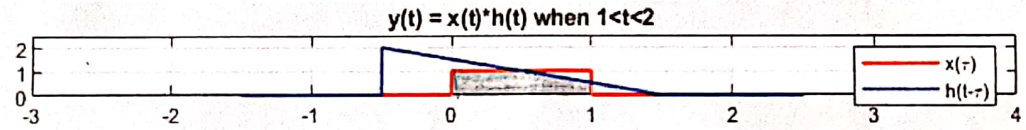
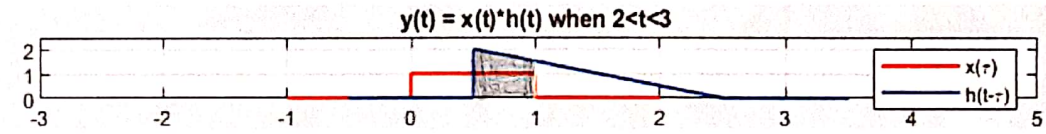
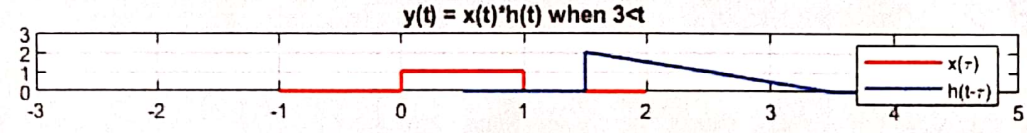
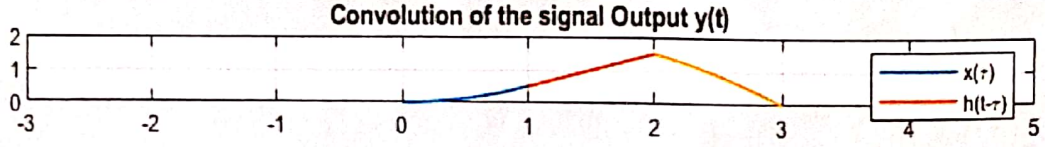










Example 2.8

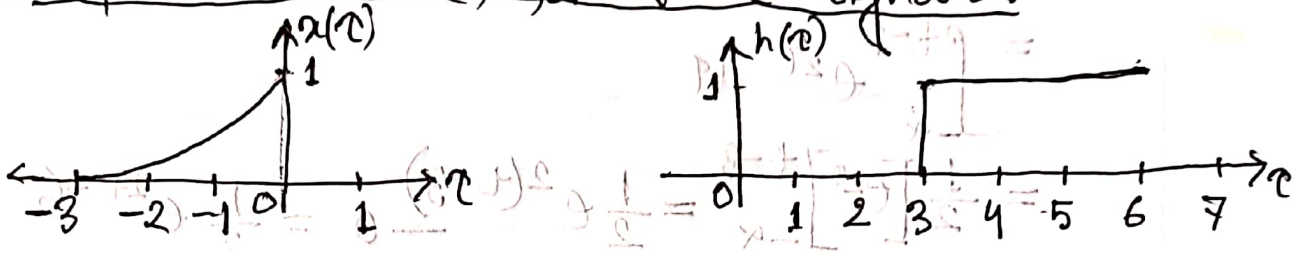
Calculation:

Input signal : $x(t) = e^{2t}.u(t)$ $\quad -2 \leq t \leq 0$

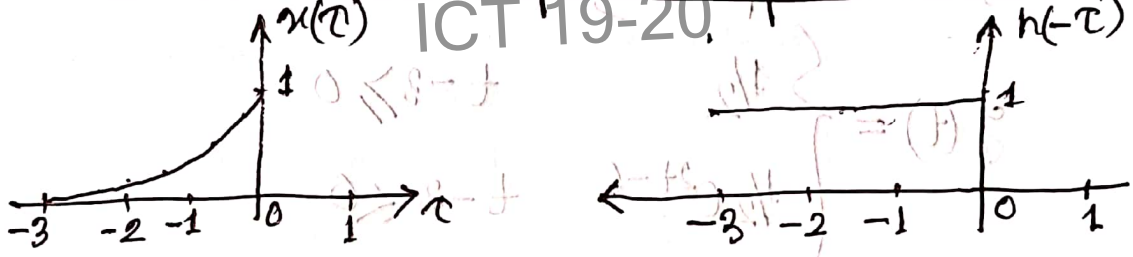
Impulse Response : $h(t) = u(t-3)$ $\quad -4 \leq t \leq 3$

Now the way of finding convolution for this signals will be started in this below:

① Replace t with (τ) for those signals:



② Time Reversal for Impulse Response $h(-\tau)$:



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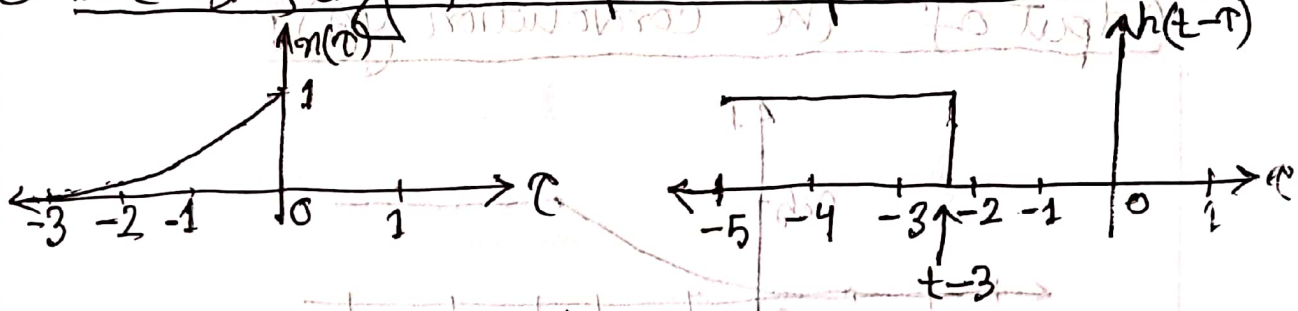
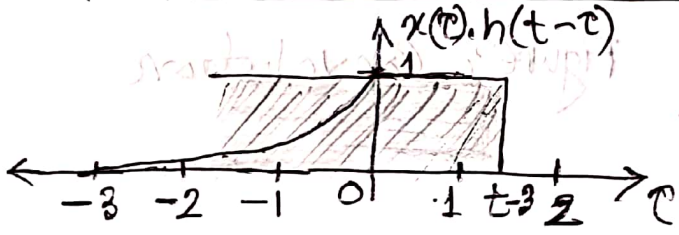
③ Time shifting for impulse Response $[h(t-\tau)]$:



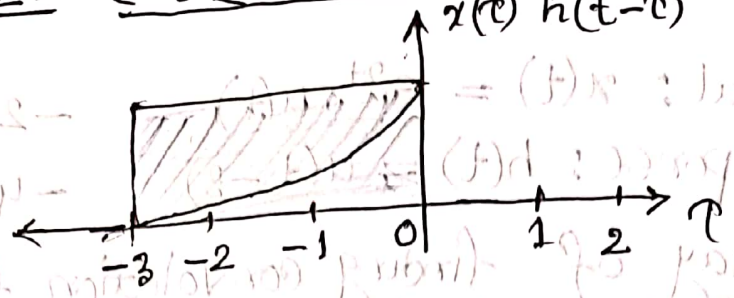
④ Case 1: $t-3 \geq 0$ and convolution starting:



$$\therefore y(t) = \int_{-\infty}^{\infty} x(\tau).h(t-\tau)\,d\tau = \int_{-\infty}^{0} e^{2\tau}\,d\tau = \frac{1}{2}\left[e^{2\tau}\right]_{-\infty}^{0} = \frac{1}{2}(1-0)$$

$$= \frac{1}{2}.$$

(ii) Case 2: $t-3 \leq 0$ and start complete overlapping



$$y(t) = \int_{-\infty}^{t-3} x(\tau)\, h(t-\tau)$$

$$= \int_{-\infty}^{t-3} e^{2\tau}\, d\tau$$

$$= \frac{1}{2}\left[e^{2\tau}\right]_{-\infty}^{t-3} = \frac{1}{2} e^{2(t-3)} - 0 = \frac{1}{2} e^{2t-6}$$

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Now, final output after convolution,

$$y(t) = \begin{cases} 1/2 & t-3 \geq 0 \\ 1/2\, e^{2t-6} & t-3 \leq 0 \end{cases}$$

Output of the convolution $y(t)$:

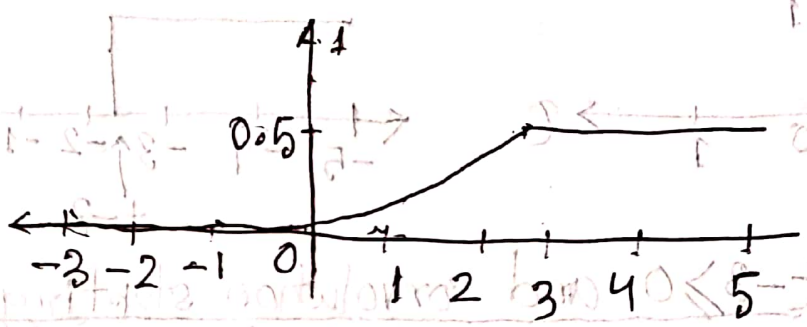
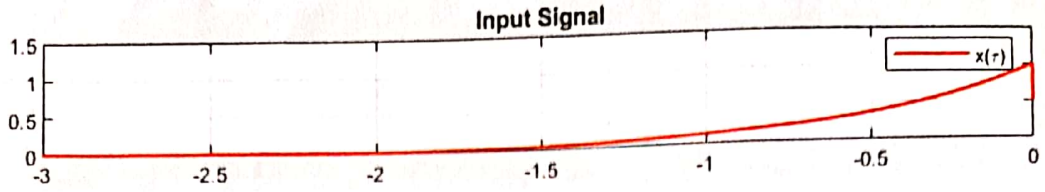
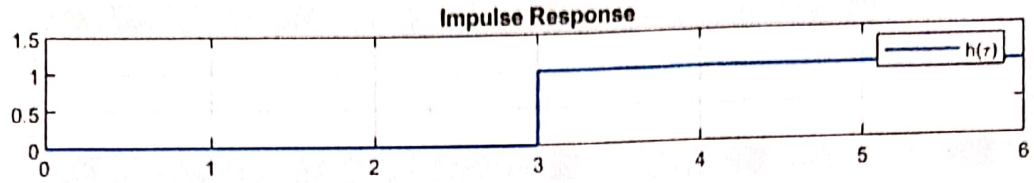
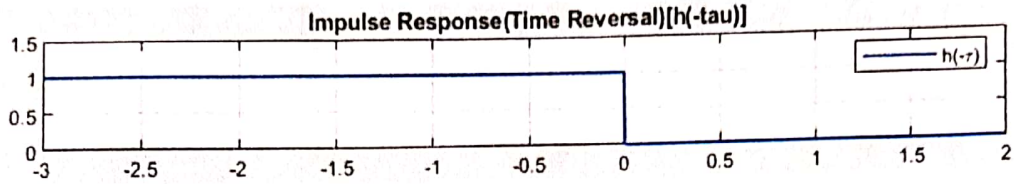


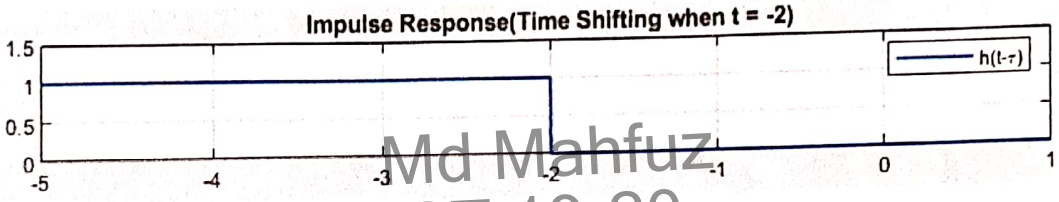
Figure: Convolution

Output:









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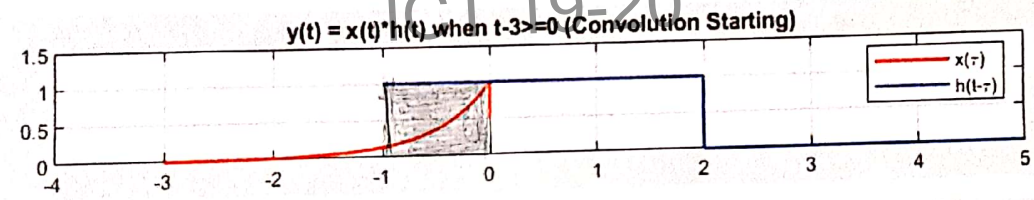
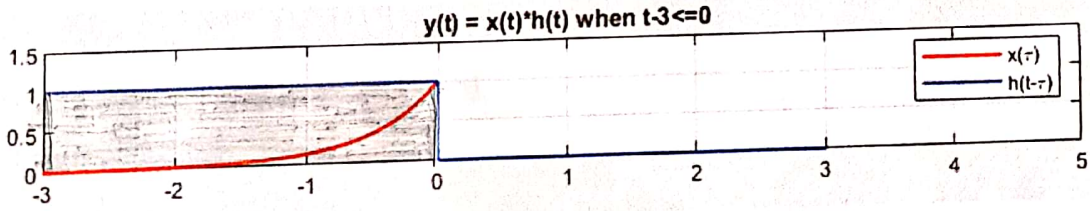
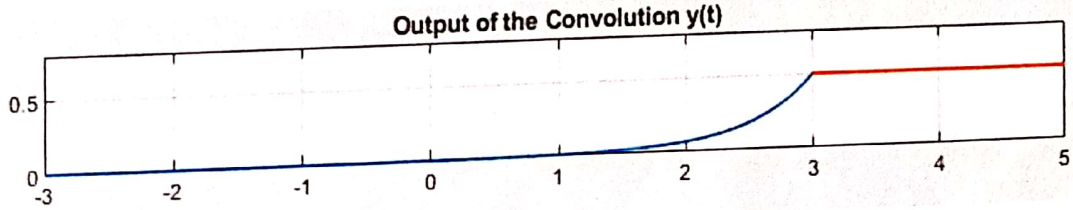






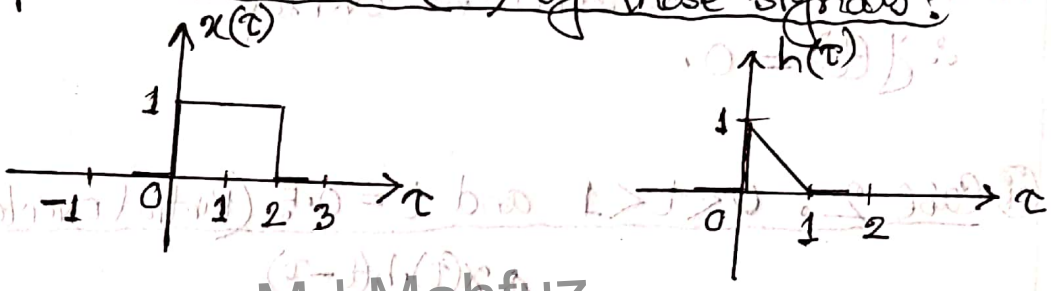
Calculation:

Let the input signal $\longrightarrow x(t) = 1, \quad 0 \leq t \leq 2$

and Impulse response $\longrightarrow h(t) = 1 - t \quad 0 \leq t \leq 1;$

Now, the way of finding convolution for this signals will be stated in this below:

① Replace t with τ (tau) by those signals:



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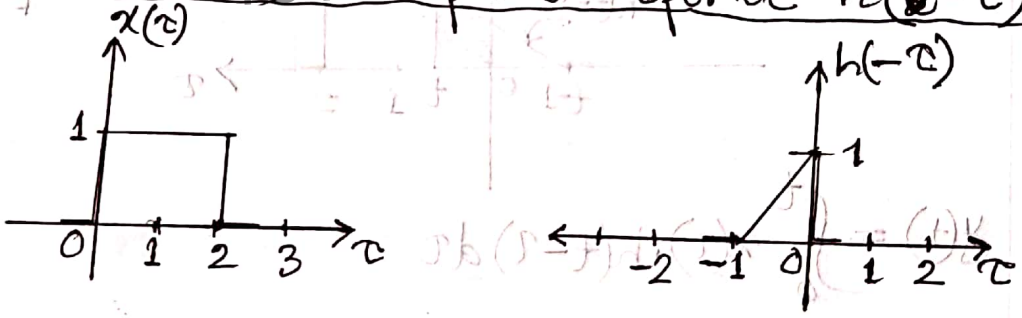
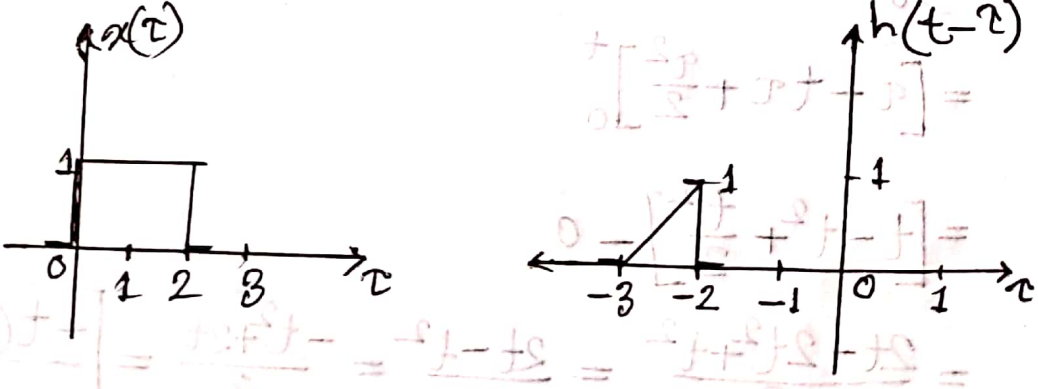
② Time Reversal for impulse response $h(-\tau)$:



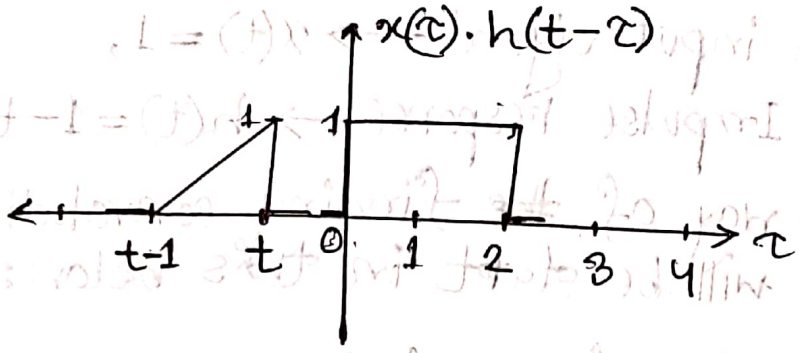
③ Time shifting for impulse response $[h(t-\tau)]$:

$t = -2$

$\therefore h(t-\tau) = h(-2-\tau)$



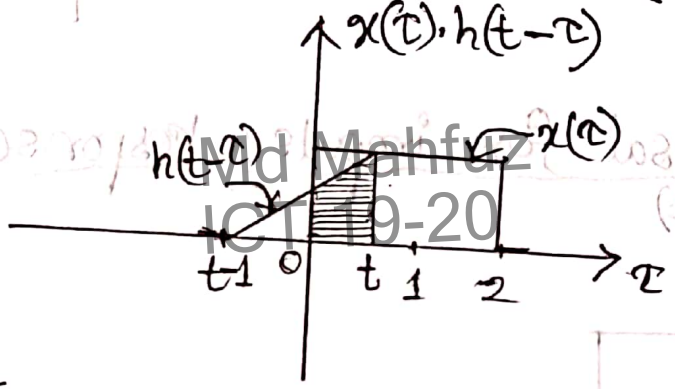
④ (i) Case 1 : $t<0$ and convolution starting:



Where, there are zero ~~ve~~ overlap.

$\therefore y(t) = 0.$

(ii) Case 2 : $0 \le t < 1$ and $t = 0.5$ (Partial overlap):



Limit : $0 \to t$

$t = 0.5$

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$$y(t) = \int_0^t x(\tau)\cdot h(t-\tau)\,d\tau$$

$$= \int_0^t 1\cdot(1-(t-\tau))\,d\tau$$

$$= \int_0^t (1-t+\tau)\,d\tau$$

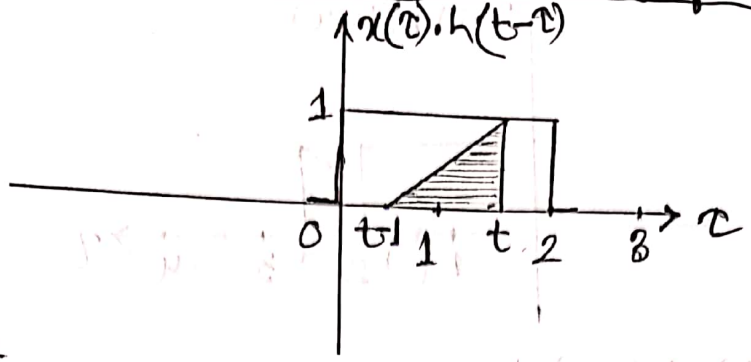
$$= \left[\tau - t\tau + \frac{\tau^2}{2}\right]_0^t$$

$$= \left[t - t^2 + \frac{t^2}{2}\right] - 0$$

$$= \frac{2t-2t^2+t^2}{2} = \frac{2t-t^2}{2} = \frac{-t^2+2t}{2} = \boxed{\frac{-t(t-2)}{2}}$$

$\therefore y(t) = (-t(t-2))/2.$

(iii) Case 3: $1<t<2$ and $t=2.5$ (Complete overlap):



Limit: $t-1 \to t$

$$y(t) = \int_{t-1}^{t} x(\tau)\cdot h(t-\tau)\,d\tau$$

$$= \int_{t-1}^{t} 1\cdot(1-(t-\tau))\,d\tau$$

$$= \int_{t-1}^{t} (1-t+\tau)\,d\tau$$

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$$= \left[\tau - t\tau + \frac{\tau^2}{2}\right]_{t-1}^{t}$$

$$= \left[t - t^2 + \frac{t^2}{2}\right] - \left[(t-1) - t(t-1) + \frac{(t-1)^2}{2}\right]$$

$$= \left[\frac{2t-2t^2+t^2}{2}\right] - \left[(t-1) - t^2 + t + \frac{t^2-2t+1}{2}\right]$$

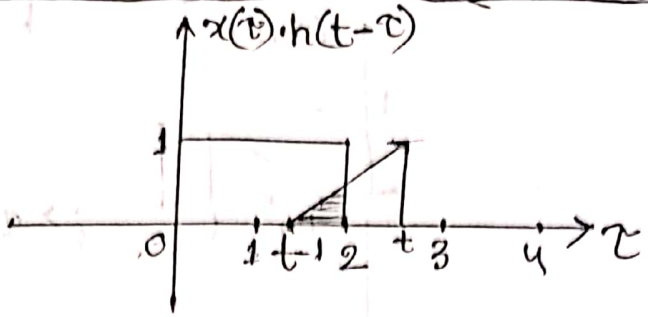
$$= \left[\frac{2t-t^2}{2}\right] - \left[2t-1-t^2+\frac{t^2-2t+1}{2}\right]$$

$$= \left[\frac{2t-t^2}{2}\right] - \left[\frac{4t-2t^2-2+t^2-2t+1}{2}\right]$$

$$= \left[\frac{2t-t^2}{2}\right] - \left[\frac{2t-t^2-1}{2}\right] = \frac{2t-t^2-2t+t^2+1}{2} = \frac{1}{2}$$

$\therefore y(t) = \frac{1}{2}$.

(N) Case 4: $2 < t < 3$ and $t = 2.5$ (Exit Partial overlap):



limit: $t-1 \to 2$

$$y(t) = \int_{t-1}^{2} x(\tau) \cdot h(t-\tau)\, d\tau$$

$$= \int_{t-1}^{2} 1 \cdot 1 \cdot (1-(t-\tau))\, d\tau$$

$$= \int_{t-1}^{2} (1-t+\tau)\, d\tau$$

$$= \left[\tau - t\tau + \frac{\tau^2}{2}\right]_{t-1}^{2}$$

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$$= \left[2 - 2t + \frac{2^2}{2}\right] - \left[\frac{2t - t^2 - 1}{2}\right]$$

$$= \frac{4-4t+4}{2} - \frac{2t+t^2+1}{2}$$

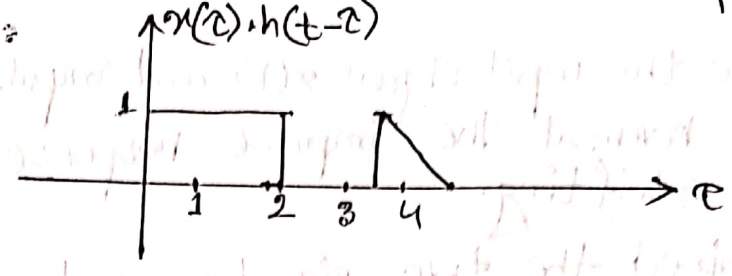
$$= \frac{4-4t+4-2t+t^2+1}{2}$$

$$= \frac{t^2-6t+9}{2} = \frac{t^2-3t-3t+9}{2} = \frac{t(t-3)-3(t-3)}{2}$$

$$= \frac{(t-3)(t-3)}{2} = \frac{(t-3)^2}{2}$$

$$\therefore y(t) = \frac{(t-3)^2}{2}$$

⑤ Case 5: $3<t<4$ and $t = 3.5$ (No overlap):



Where,
there were no overlap
$\therefore y(t) = 0$

Now, final output after convolution:

$$y(t) = \begin{cases} 0 \longrightarrow & t<0 \\ -(t(t-2))/2 \rightarrow & 0<t<1 \\ 1/2 \longrightarrow & 1<t<3 \\ (t-3)^2/2 \longrightarrow & 2<t<3 \\ 0 \longrightarrow & 3<t<4. \end{cases}$$

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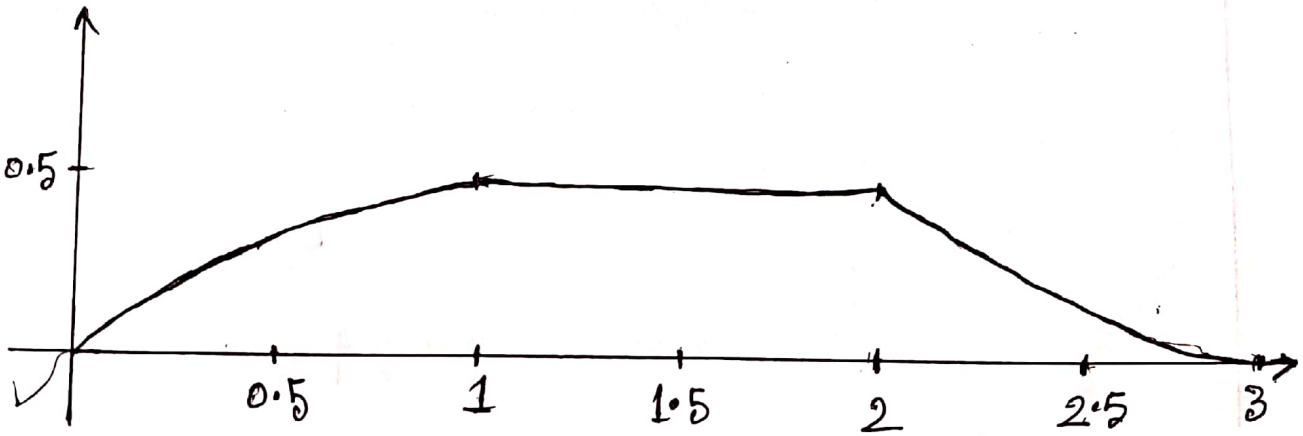
Convolution of the signals output:



Algorithm:

(i) Start

(ii) Define the input signal $x(t)$ and impulse response $h(t)$.

(iii) Time reversal the impulse response and then time shifting.

(iv) Combined the two signals and pass through the impulse response of the limit minus infinity to plus infinity

(v) The signals are continuous time signal (Integration the signals) or Discrete time signals (Summation the signals).

(vi) Plotting the combined signal respect to the time

(vii) Stop.

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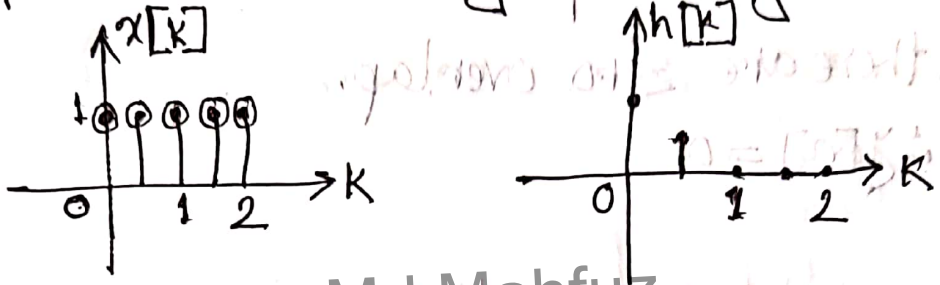
For Discrete time Convolution sum: (Calculation)

Input signal, $x[n] = 1$ $\quad 0 \le n \le 2$;

Impulse Response, $h[n] = 1-n$ $\quad 0 \le n \le 2$;

Now, the way of finding convolution for this signal will be shown in below:

① Replace n with k by those signals:



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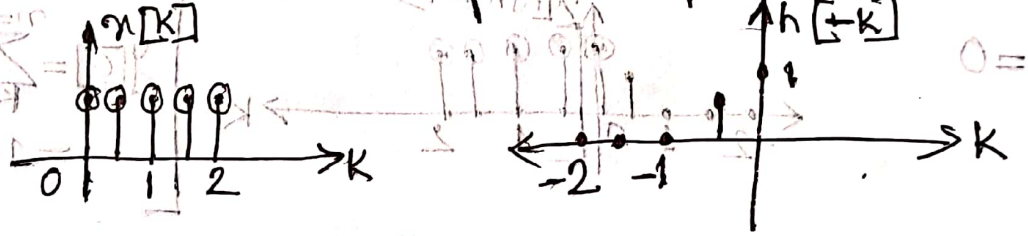
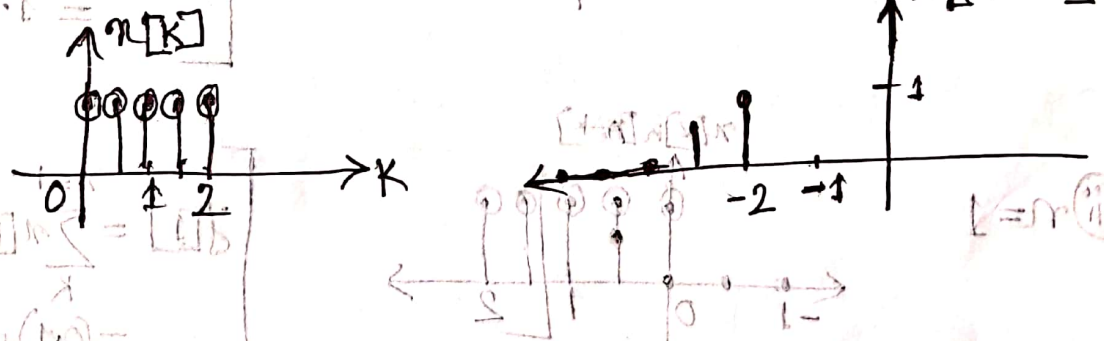
② Time Reversal for impulse response → $h[-k]$:



③ Time shifting for impulse response $h[n-k]$

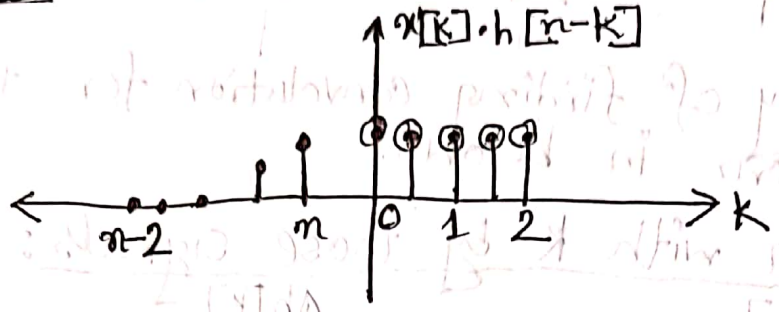
$n = -2$

$\therefore h[n-k] = h[-2-k]$



Now, start convolution to passing through the impulse response signal $-\infty$ to $+\infty$.

Case 1: $n < 0$



Here, there are zero overlap.

$\therefore y[n] = 0$

Case 2: $(0 \leq n \leq 2)$

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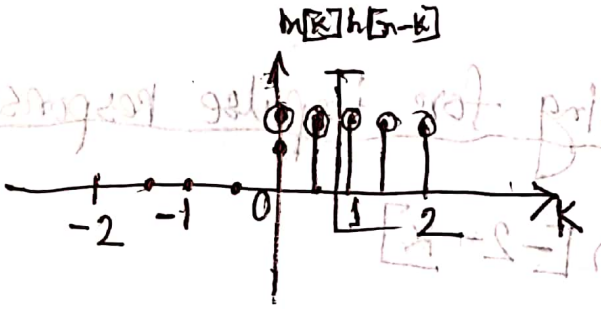
(i) $n = 0$

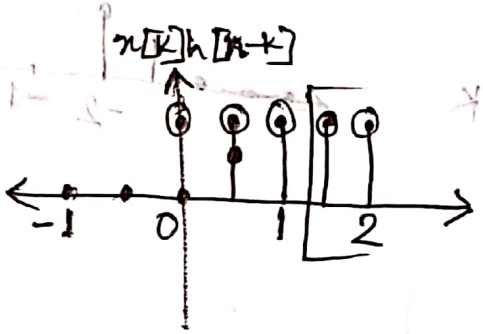
$$y[0] = \sum_{k}^{n=0} x[k] \cdot h[n-k]$$
$$= 1 \cdot 1 = 1$$

(ii) $n = 0.5$



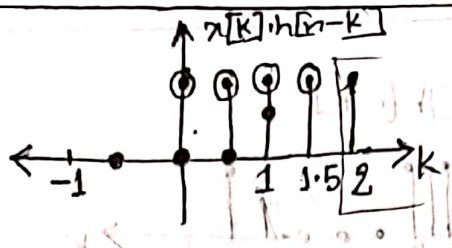
$$y[0.5] = \sum_{k}^{n=0.5} x[k]\, h[0.5-$$
$$= (0.5 \times 1) + (1 \times 1)$$
$$= 1.5$$

(iii) $n = 1$



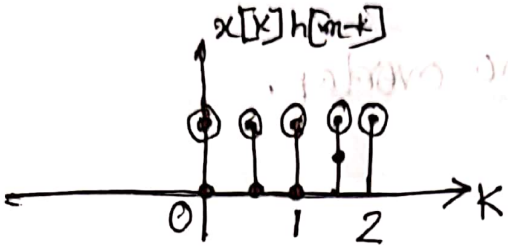
$$y[1] = \sum_{k}^{n=1} x[k]\, h[1-k]$$
$$= (0 \times 1) + (0.5 \times 1) + (1 \times 1)$$
$$= 1.5$$

(iv) $n = 1.5$



$$y[1.5] = \sum_{k}^{n=1.5} x[k]\, h[1.5-k]$$
$$= (0\times1)+(0\times1)+(0.5\times1)+(1\times1)$$
$$= 1.5$$

(v) $n = 2$



$$y[2] = \sum_{k}^{n=2} x[k]\, h[2-k]$$
$$= (0\times1)+(0\times1)+(0\times1)+(0.5\times1)+(1\times1)$$
$$= 1.5$$

Case 3: $(2 < n \le 4)$

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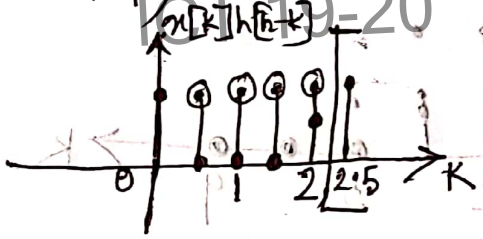
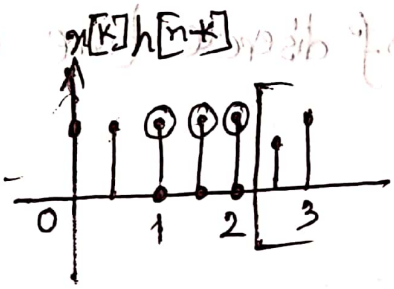
(i) $n = 2.5$



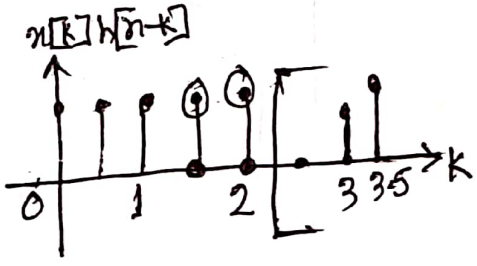
$$y[2.5] = \sum_{k}^{n=2.5} x[k]\, h[2.5-k]$$
$$= (0\times1)+(0\times1)+(0\times1)+(0.5\times1)$$
$$= 0.5$$

(ii) $n = 3$



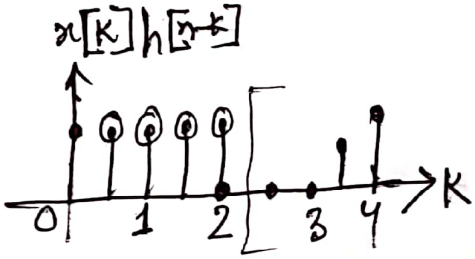
$$y[3] = \sum_{k}^{n=3} x[k]\, h[3-k]$$
$$= (0\times1)+(0\times1)+(0\times1)$$
$$= 0$$

(iii) $n = 3.5$



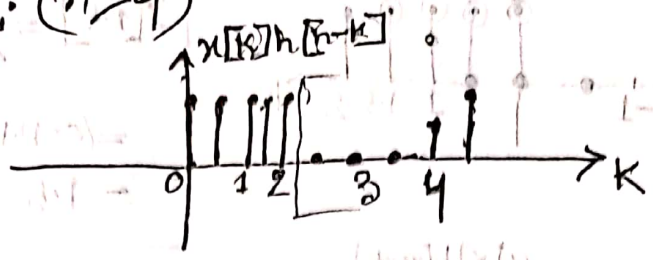
$$y[3.5] = \sum_{k}^{n=3.5} x[k]\, h[3.5-k]$$
$$= (0\times1)+(0\times1) = 0$$

(iv) $n = 4$



$$y[4] = \sum_{k}^{n=4} x[k]\, h[4-k]$$
$$= (0\times1) = 0$$

Case 4: $(n>4)$



Here, there are no overlap.

$\therefore y[n] = 0$

$\therefore y[n] = [1 \quad 1.5 \quad 1.5 \quad 1.5 \quad 1.5 \quad 0.5 \quad 0 \quad 0 \quad 0]$

Output of the convolution:

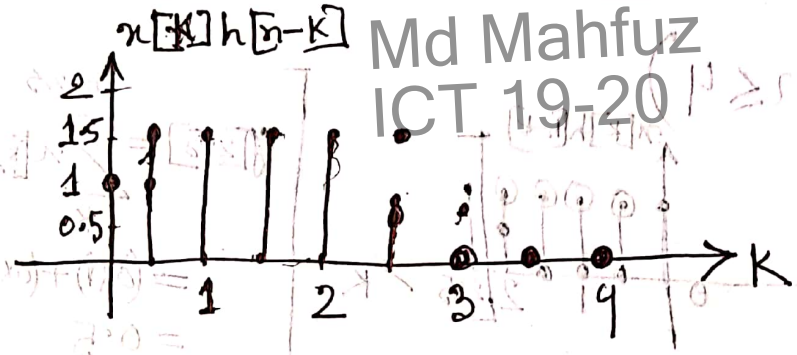


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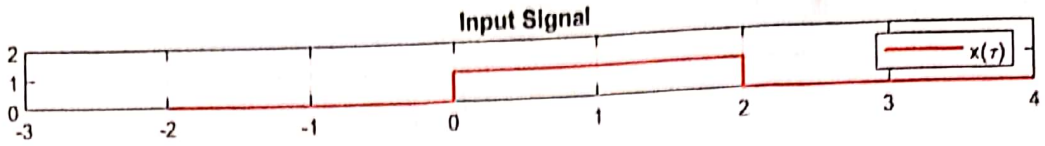
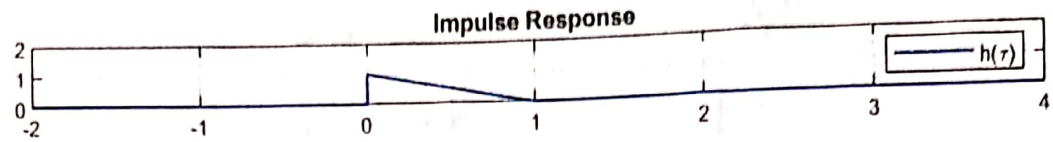
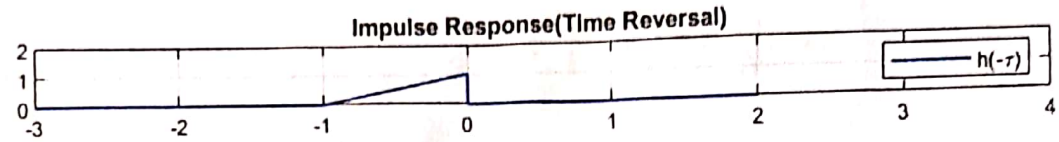
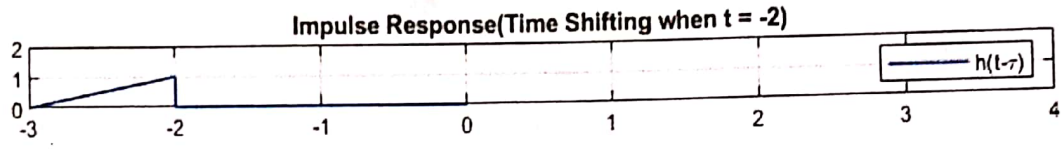
Figure: Output of discrete time convolution

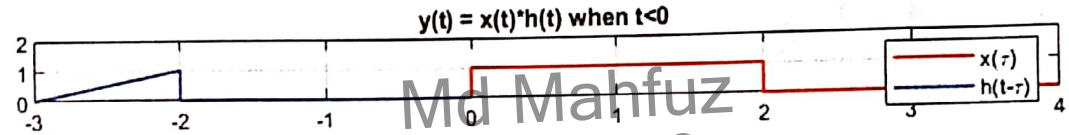
Output:











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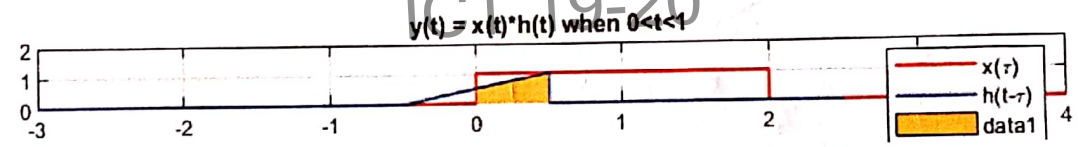
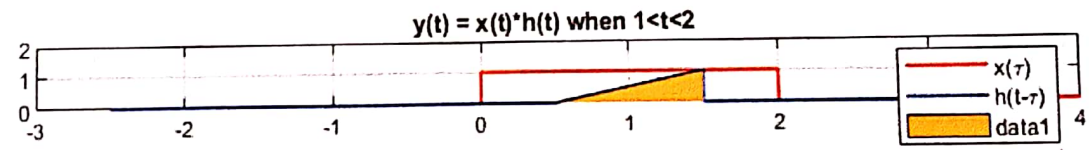
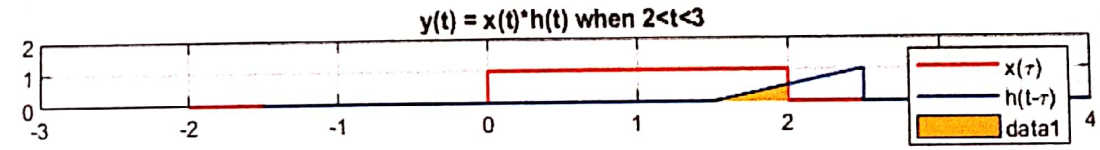
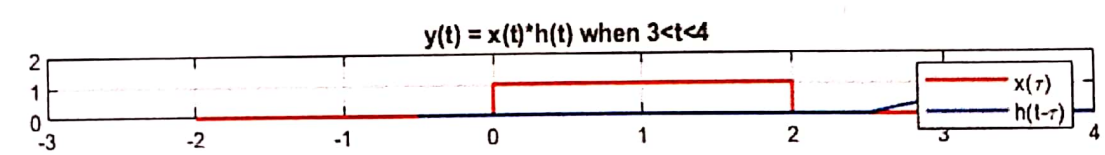
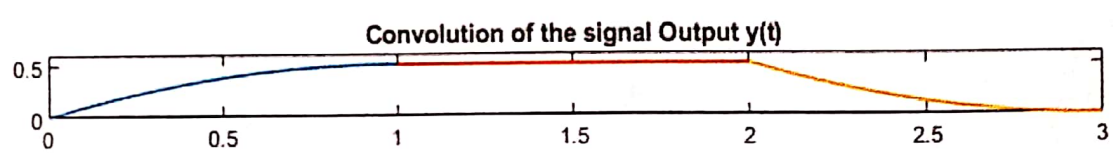


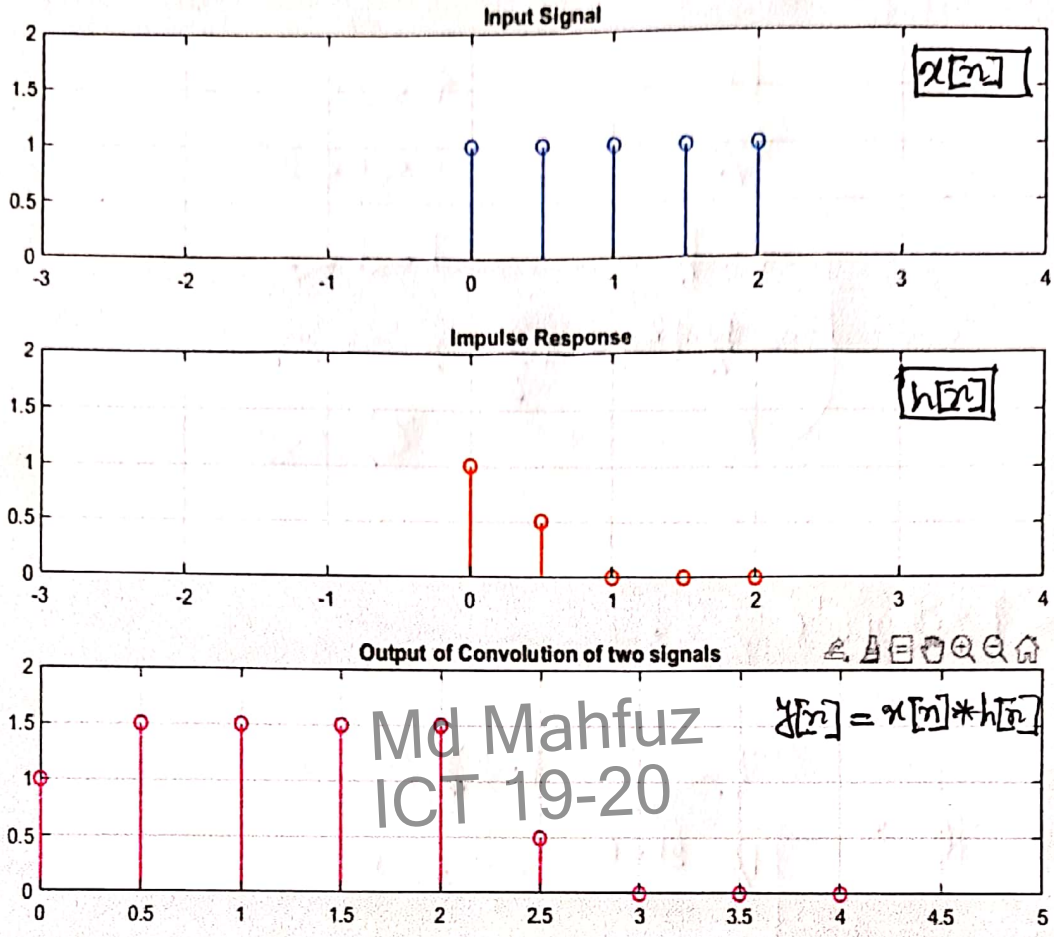








Output of Discrete Time Convolution:



Result and Discussion:

From the above figure, we see that, the convolution of the two signals about input signals and impulse response,

In case 1, there are no overlap, so the integration of this case is equal to zero. ($y(t)=0$)

In case 2, signals are initial partial overlap, so the integration of this area, $y(t)=(-t(t-2))/2$.

In case 3, signals are complete overlap, so the integration of this module, $y(t)=1/2$.

In case 4, signals are exit partial overlap, so the integration of the area, $y(t)=((t-3)^2/2)$.

In Case 5, when $t>2$, there are no overlap, so the integration of this module, $y(t)=0$

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For discrete time convolution,

$y[n]=[1 \quad 1.5 \quad 1.5 \quad 1.5 \quad 1.5 \quad 0.5 \quad 0 \quad 0 \quad 0]$

For, Example 2.6,

$$y(t)=\begin{cases}0 & t<0\\ 1-e^{-t} & t>0\end{cases}$$

For, Example 2.7,

$$y(t) = \begin{cases} 0 & t<0 \\ t^2/2 & 0<t<1 \\ t-1/2 & 1<t<2 \\ -1/2\,t^2+t+\frac{3}{2} & 2<t<3 \\ 0 & 3<t \end{cases}$$

For, Example 2.8,

$$y(t) = \begin{cases} 1/2 & t+3 \geq 0 \\ 1/2\,e^{2t+6} & t-3 \leq 0 \end{cases}$$

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Experiment No: 06

Name of the experiment: Write a program to find the trigonometric Fourier series coefficient of rectangular periodic signal. Reconstruct the signal by combining the Fourier series coefficient with appropiate weightings.

AIM

To find the trigonometric series coefficients of rectangular periodic signal and reconstruct the signal.

Objectives:

After compliting this experiment, we are able to know about this:

(i) Determine the Fourier series coefficient of rectangular periodic signal.

(ii) Know about rectangular pulse for periodic signal.

(iii) Implement the course concept on using MATLAB.

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Theory:

A periodic signal can be represented over a certain interval of time in terms of the linear combination of orthogonal functions, if these orthogonal function are trigonometric function, then the Fourier series represention is known as trigonometric Fourier series.

Explanation:

Consider a sinusoidal signal $x(t) = A\sin\omega_0 t$ which is periodic with time period T such that $T = 2\pi/\omega_0$.

If the frequencies of two sinusoids are integral multiples of a fundamental frequency (ω_0), then the sum of these two sinusoids is also periodic

Let the signal $x(t)$ that the sum of sine and cosine functions is given by —

$$x(t) = a_0 + a_1 \cos \omega_0 t + a_2 \cos 2\omega_0 t + a_3 \cos 3\omega_0 t + \cdots + a_k \cos k\omega_0 t + b_1 \sin \omega_0 t + b_2 \sin 2\omega_0 t + b_3 \sin 3\omega_0 t + \cdots + b_k \sin k\omega_0 t$$

$$\Rightarrow x(t) = a_0 + \sum_{n=1}^{K} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \quad \text{—— (i)}$$

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Where, $a_0, a_1, a_2 \cdots a_k$ and $b_0, b_1, b_2 \cdots b_k$ are constant and ω_0 is the fundamental frequency.

Again, if a signal $x(t)$ is a periodic signal, then it must satisfy the following condition —

$x(t) = x(t+T)$: for all t

Therefore,

$$\Rightarrow x(t+T) = a_0 + \sum_{n=1}^{K} a_n \cos n\omega_0 (t+T) + b_n \sin n\omega_0 (t+T)$$

$\because$ Time period, $T = \left(\frac{2\pi}{\omega_0}\right)$

$$\therefore \Rightarrow x(t+T) = a_0 + \sum_{n=1}^{K} a_n \cos n\omega_0 \left(t + \frac{2\pi}{\omega_0}\right) + b_n \sin n\omega_0 \left(t + \frac{2\pi}{\omega_0}\right)$$

$$\Rightarrow x(t+T) = a_0 + \sum_{n=1}^{K} a_n \cos (n\omega_0 t + 2n\pi) + b_n \sin (n\omega_0 t + 2n\pi)$$

$\because \cos(2n\pi + \theta) = \cos\theta$ and $\sin(2n\pi + \theta) = \sin\theta$

So,

$$x(t+T) = a_0 + \sum_{n=1}^{K} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) = x(t)$$

$\therefore$ —— (ii)

Hence, the trigonometric form of Fourier series can be defined as under—

The infinite series of sine and cosine terms of frequencies $0, \omega_0, 2\omega_0 \dots k\omega_0$ is called the trigonometric form of Fourier series and can be represented as,

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Where, a_0, a_n, b_n are called trigonometric Fourier series coefficients.

$$a_0 = \frac{1}{T_0}\int_{T_0} x(t)\,dt \quad \text{and} \quad a_n = \frac{2}{T_0}\int_{T_0} x(t)\cos n\omega_0 t\,dt$$

$$b_n = \frac{2}{T}\int_{T_0} x(t)\sin n\omega_0 t\,dt$$

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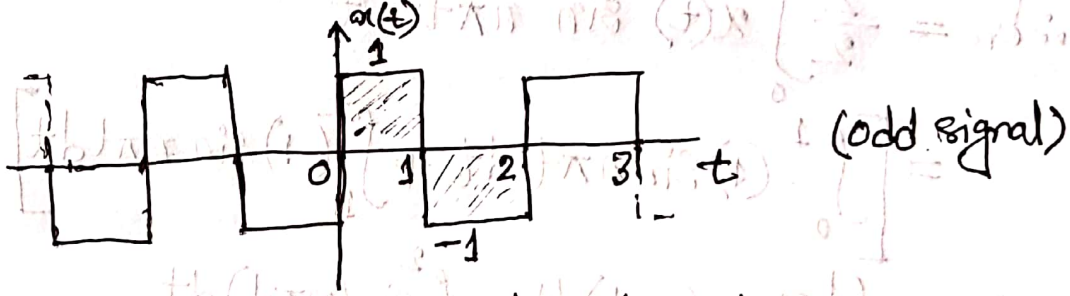
Calculation:

Let a trigonometric Fourier series rectangular pulse—

$$x(t) = \begin{cases} 1, & -1 < t < 1 \\ 0 & \text{otherwise} \end{cases}$$



(odd signal)

Figure: A rectangular pulse.

To ~~find~~ analysis, in the given signal $x(t)$ is symmetrical about the t-axis, this means $a_0 = 0$.

Now, we are also seen that the signal is purely a odd signal and there are no cosine term here this means, $a_n = 0$.

Thus, the signal is purely odd signal that means only sine term are present.

So, we rewritten the trigonometric Fourier series signal as $x(t) = \sum_{n=1}^{\infty} b_n \sin \omega_0 t \, dt$

Now, we are calculating the coefficient of rectangular pulse for trigonometric Fourier series,

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$$b_n = \frac{2}{T_0}\int_{T_0} x(t) \sin n\omega_0 t \, dt$$

From the following figure of the signal, every cycle is completed $T_0 = (2-0) = 2$ sec in the periodic sig

$$\therefore \omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{2} = \pi \text{ rad/sec}$$

$$\therefore b_n = \frac{2}{2}\int_0^2 x(t) \sin n\pi t \, dt$$

$$= \left[\int_0^1 (1) \sin n\pi t \, dt + \int_1^2 (-1) \sin n\pi t \, dt\right]$$

$$= \int_0^1 \sin(n\pi t) dt - \int_1^2 \sin(n\pi t) dt$$

$$= -\left[\frac{\cos(n\pi t)}{n\pi}\right]_0^1 - \left[-\frac{\cos(n\pi t)}{n\pi}\right]_1^2$$

$$= -\left[\frac{\cos(n\pi) - 1}{n\pi}\right] + \left[\frac{\cos(n\pi) - \cos(n\pi)}{n\pi}\right]$$

$$\frac{-\cos(n\pi)+1}{n\pi} + \frac{\cos(2n\pi)-\cos(n\pi)}{n\pi}$$

$$= \frac{1}{n\pi}\left[\cos(2n\pi) - 2\cos(n\pi) + 1\right]$$

Now, if n = even (2, 4, 6, 8 - ...), then,

$b_n = 0$

and, if n = odd (1, 3, 5, 6 ... ...), then

$$b_n = \frac{4}{n\pi}$$

$$\therefore x(t) = \sum_{n=1}^{\infty} b_n \sin\omega_0 t \, dt$$

$$= \frac{4}{\pi}\sin\pi t + \frac{4}{3\pi}\sin 3\pi t + \frac{4}{5\pi}\sin 5\pi t + \text{----}$$

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Algorithm:

(i) Start

(ii) Input values for $f_s = 100$ and $T = 2$ taken.

(iii) Set the ploting limit of the rectangular pulse.

$K = -6 : 1/fs : 6 - 1/fs$

(iv) Integrate the fourier series coefficient to determine the reconstruction of the signal.

(v) Plot the appoximation of the signal

(vi) Plot two signal together and compare the signal how much different it

(vii) Stop.

Result and discussion:

In this experiment, the signal of the rectangular pulse is a odd signal that is the coefficient of the signal ~~a_0, b_n~~ $a_0, a_n = 0$.

If the signal is a odd signal that is the coefficient b_n is fourth, from the calculation, we have

If n = even, $b_n = 0$ and

n = odd, $b_n = \frac{4}{n\pi}$

To combine the all coefficient, we are reconstructing the main signal

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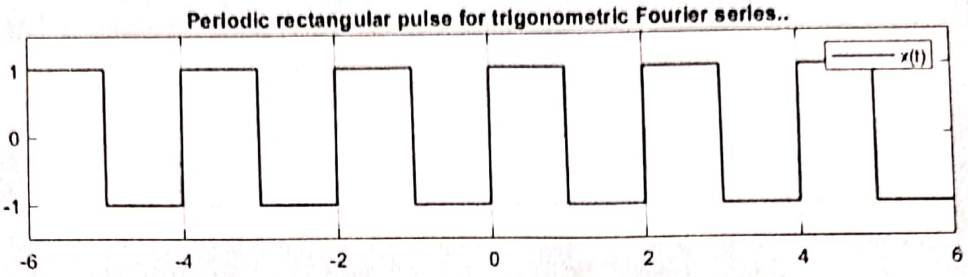
$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

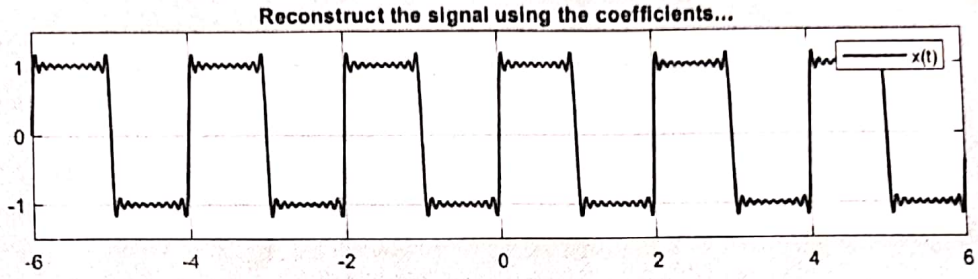
$$= 0 + 0 + \frac{4}{n\pi} \sin n\pi t$$

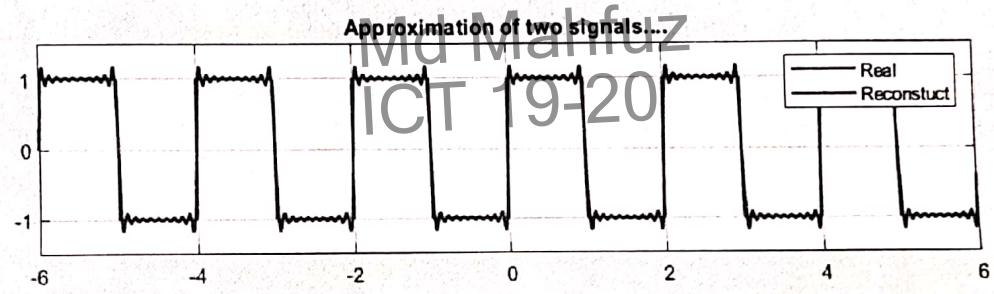
$$= \frac{4}{n\pi} \sin n\pi t$$

$$= \frac{4}{\pi} \sin \pi t + \frac{4}{3\pi} \sin 3\pi t + \cdots$$

Output:







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Experiment No: 07

Experiment Name: Write a program to find exponential Fourier series coefficients of a periodic rectangular signal. Plot the discrete spectrum of the signal.

AIM:

To find the exponential Fourier series coefficient of a rectangular periodic signal and plot discrete time spectrum.

Objectives:

After complete this experiment, we are able to know about this.

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(i) Determine the Fourier series coefficients of rectangular periodic signal.

(ii) Know about rectangular pulse for periodic signal.

(iii) Implement the course concepts on using MATLAB.

Theory:

Exponential Fourier series: A periodic signal can be represented over a certain interval of time in terms of linear combination of orthogonal functions. If these orthogonal functions are exponential functions, then it is called the exponential Fourier series.

For any periodic signal $x(t)$, the exponential form of Fourier series is given by,

$$x(t) = \sum_{n=-\infty}^{\infty} C_k e^{jk\omega_0 t} \quad \text{---------} (1)$$

Coefficients of Exponential Fourier Series

From equation (i), we multiply this by $e^{jk\omega_0 t}$, we obtain,

$$x(t) \cdot e^{-jk\omega_0 t} = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \cdot e^{-jk\omega_0 t}$$

Integrating both sides from 0 to $T = 2\pi/\omega_0$, we have,

$$\int_0^T x(t) \cdot e^{-jk\omega_0 t} dt = \int_0^T \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \cdot e^{-jk\omega_0 t} dt$$

Here, T is the fundamental period of x(t), and consequently, we are integrating over period.

$$\int_0^T x(t) \cdot e^{-jk\omega_0 t} dt = \sum_{n=-\infty}^{\infty} c_n \left[\int_0^T e^{j(n-k)\omega_0 t} dt \right] \quad \text{(ii)}$$

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The evaluation of the bracked integral is straightforward, re-written this integral using Euler's formula we obtain,

$$\int_0^T e^{j(k-n)\omega_0 t} dt = \int_0^T \cos(k-n)\omega_0 t \, dt + j\int_0^T \sin(k-n)\omega_0 t \, dt \quad \text{(iii)}$$

(1) For $k \neq n$, both the integrals on the right-hand side of equation (ii) are zero.

(2) Again, $k = n$, the integrad on the left hand side of equation (iii) $\longrightarrow$ equals to 1, and thus the integrals $\longrightarrow$ equals to T.

$$\int_0^T e^{j(k-n)\omega_0 t} dt = \begin{cases} T, & k = n \\ 0, & k \neq n \end{cases}$$

Now, putting this value of the equation (ii), we have,

$$\int_0^T x(t) e^{-jn\omega_0 t} dt = T \cdot c_n$$

$$\therefore \boxed{c_k = \frac{1}{T}\int_0^T x(t) e^{-jk\omega_0 t} dt}$$

Which provides the equation for determining the coefficients.

Calculation:

Consider a periodic rectangular signal, ——

$$x(t) = \begin{cases} 1, & -1 < t < 1 \\ 0 & \text{otherwise} \end{cases}$$

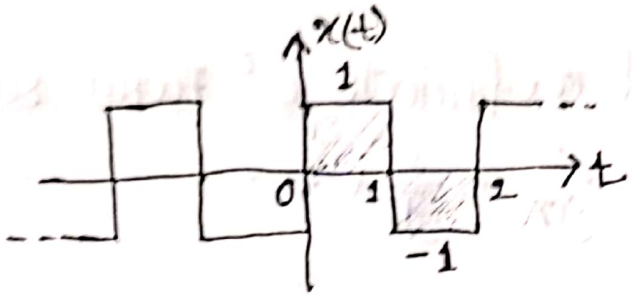

Figure: rectangular pulse

To determine the coefficients of the signal, we know that,

$$C_n = \frac{1}{T}\int_{T_0} x(t)\cdot e^{-jn\omega_0 t}$$

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[Where, $T = 2\text{sec}$

$\therefore \omega_0 = \frac{2\pi}{T} = \frac{2\pi}{T} = \pi$

$$= \frac{1}{2}\left[\int_0^1 (1)\cdot e^{-jn\pi t}\,dt + \int_1^2 (-1)\cdot e^{-jn\pi t}\,dt\right]$$

$$= \frac{1}{2}\left[\int_0^1 e^{-jn\pi t}\,dt - \int_1^2 e^{-jn\pi t}\,dt\right]$$

$$= \frac{1}{2}\left\{\left[\frac{e^{-jn\pi t}}{-jn\pi}\right]_0^1 - \left[\frac{e^{-jn\pi t}}{-jn\pi}\right]_1^2\right\}$$

$$= \frac{1}{2}\left\{\left[\frac{e^{-jn\pi} - 1}{-jn\pi}\right] - \left[\frac{e^{-j2n\pi} - e^{-jn\pi}}{-jn\pi}\right]\right\}$$

$$= \frac{1}{-2jn\pi}\left\{\left[(-1)^n - 1\right] - \left[1 - (-1)^n\right]\right\}$$

$$= \frac{1}{-2jn\pi}\left[(-1)^n - 1 - 1 + (-1)^n\right]$$

[$e^{j\pi} = -1$
$(e^{j\pi})^2 = (-1)^n$
$e^{j2\pi} = 1$
$(e^{j\pi})^n = (-1)^n$

When, n = even (2, 4, 6, 8 ---)

$$C_n = \frac{1}{-2jn\pi}\left[1 - 1 - (1 - 1)\right] = 0$$

When, n = ~~even~~ odd (1, 3, 5, 7, ---)

$$c_n = \frac{1}{-2jn\pi}\left([(-1)-1]-[1-(-1)]\right) = \frac{1}{-2jn\pi}[-2-[2]] = \frac{1}{-2jn\pi} \times (-4$$

$$= \frac{2}{jn\pi}$$

∴ The exponential coefficients of Fourier series,

$$c_n = \text{-----} \frac{2}{j\pi} + \frac{2}{j\pi} + \text{----}$$

$$\therefore x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

$$= \text{-----} + \frac{2}{j\pi} e^{j\omega_0 t} + \frac{2}{j\pi} e^{j\omega_0 t} + \text{----}$$

Algorithm:

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(i) Start

(ii) Input values for fs = 100 and T = 2 taken.

(iii) Set the ploting limit of the signal of rectangular pul
K = −6 : (1/fs) : 6 − 1/fs.

(iv) Integrate the fourier series coefficient to determ the original signal.

(v) To determine the discrete spectrum of the original signal using ~~mat~~ matlan fft() function and store the value into a variable (Y).

(vi) Plot the absulate value of the Y to using MATLAB Library function abs(Y).

(vii) Stop.

Result and Discussion:

From the experiment,
the coefficient of exponential Fourier series, we have

$C_n = \cdots\cdots, -\frac{2}{j3\pi}, \frac{2}{j\pi}, \frac{2}{j3\pi}, \cdots$

The absulate value of the signal

$Y = fft(x(t))$

and absulate value of the signal $= abs(Y)$.

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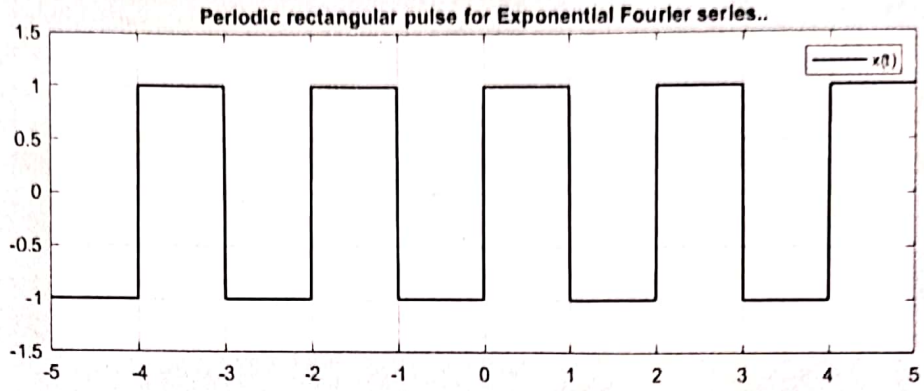
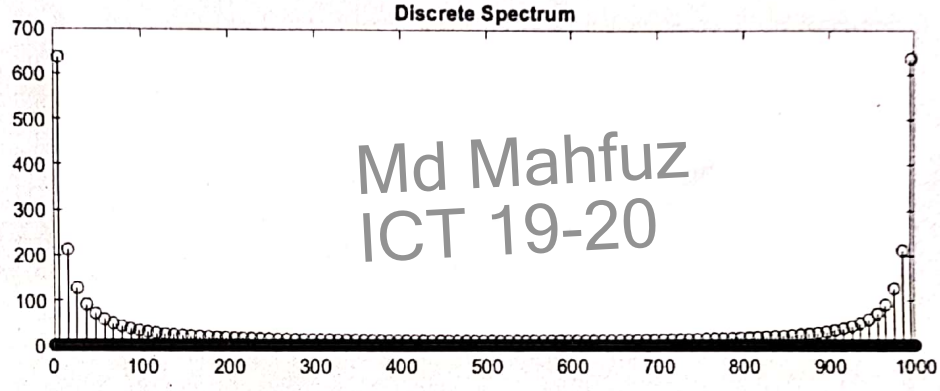
Output:





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Experiment No: 08

Name of the Experiment: Write a program to perform transformations of signals into time and frequency domain

AIM:

To know about the frequency and time domain of Fourier Transform.

Objectives:

(i) Under about Fourier Transform over Fourier series.
(ii) How to work time & frequency domain in Fourier Transform.

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Theory:

The main drawback of Fourier series is, it is only applicable to periodic signals. ~~They~~ There are some naturally produced signals such as non periodic or aperiodic, which we can not represent using Fourier series. To overcame this shortcoming, Fourier developed a mathematical model to transform signals between time domain to frequency domain.

Basic Fourier Transform pair:

Time domain	Frequency domain
$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega)\, e^{j\Omega t} d\Omega$	$X(\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t} dt$

$x(t)$ can be reconstruct from its spectrum using the inverse Fourier Transform.

The above two equation are ~~beff~~ referred to as the Fourier Transform Pair with the first one being Synthesis equation (time domain) and the second one being the analysis equation (frequency domain).

Notation

$$X(j\omega) = F\{x(t)\}$$
$$x(t) = F^{-1}\{X(j\omega)\}$$

$x(t)$ and $X(j\omega)$ form a Fourier transform pair, denoted by

$$x(t) \overset{FT}{\rightleftharpoons} X(j\omega)$$

The Fourier transform is primary tool for analyzing signals and signals and processing systems in frequency domain, especially when signals are sampled or converted from discrete time to continuous time

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Calculation:

Consider a time domain signal, $x(t) = e^{-at} \cdot u(t)$

The Fourier Trans form of $x(t)$ is ——

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t}\, dt$$

$$= 0 + \int_{0}^{\infty} e^{-at} \cdot e^{-j\omega t}\, dt \qquad [\because u(t) \Rightarrow (t>=0)]$$

$$= \int_{0}^{\infty} e^{-t(a+j\omega)}\, dt$$

$$= \left[\frac{e^{-t(a+j\omega)}}{-(a+j\omega)}\right]_0^{\infty}$$

$$= \left[\frac{0-1}{-(a+j\omega)}\right] = \frac{1}{a+j\omega}$$

%% a=1 to plot the frequency domain

$$\therefore x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega).e^{j\omega t}\,d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{a+j\omega}.e^{j\omega t}\,d\omega$$

$$= FT^{-1}\{X(j\omega)\}$$

Algorithm:

(i) Start

(ii) Define a time domain signal $x(t) = e^{-at}.u(t)$

(iii) Plot the time domain signal into a time interval.

(iv) Use matlab syms t to integration the time domain signal.

(v) To transformation the time domain signal to frequency domain signal use fourier() library function as Y = fourier(x(t)).

(vi) Store the frequency domain signal into Y.

(vii) Plot the frequency domain signal.

(viii) To back frequency domain signal to time domain signal use ifourier() library function. Such as ifourier(Y).

(ix) Plot the time domain signal.

(x) Stop.

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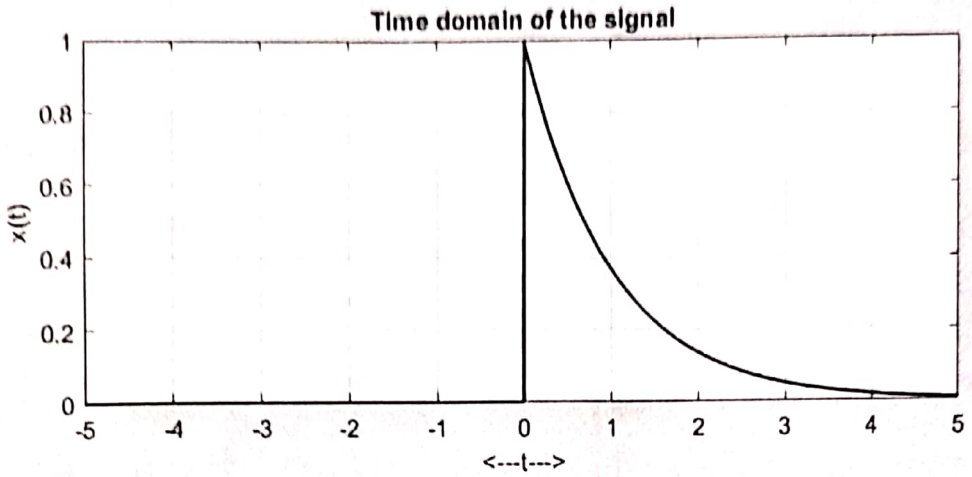
Output:



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Experiment No: 09

Name of the experiment: Find the Fourier Transform of a square pulse. Plot its amplitude and phase spectrum.

AIM:

To know about amplitude and phase spectrum of a square pulse of Fourier Transform.

Objectives:

(i) Know about Fourier Transform over Fourier series.
(ii) Know about phase amplitude and phase spectrum.
(iii) Implement the phase spectrum and amplitude spectrum using MATLAB.

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Theory:

The Fourier Transform describes a way of decomposing a function into some orthogonal basis functions in just the same way as we decompose a point in time to frequency domain.

The main drawback of Fourier series is, it is only applicable to periodic signals. There are some naturally produces signal as non-periodic or aperiodic which we can not represent using Fourier series. To overcome this short coming, Fourier developed a mathematical tool to transform signal between time domain to frequency domain.

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega \quad \text{and} \quad X(j\omega) = \int_{-\infty}^{\infty} x(t)\cdot e^{-j\omega t}\, dt$$

A square wave: A square wave is a non periodic sinusoid periodic waveform in which the amplitude alternates at a steady frequency between fixed minimum and maximum are instantaneous.
The ratio of the high period to the the total period of a pulse wave is called the duty ~~ele~~ cycle. A tr square wave have a 50% duty ~~ele~~ cycle.

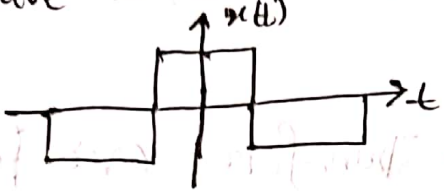

Figure: a square wave.

The amplitude and phase spectrum.

The signal spectrum specify the amplitude and phase spectrum. The amplitude spectrum specifies the amplitude of a signal components as a function of component frequency.

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Amplitude: The amplitude of a non-periodic signal i its magnitude compared with a reference value.

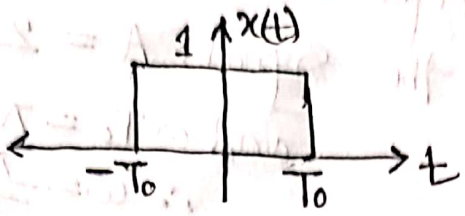
and phase spectrum specifies the phase of signal components as a function of components frequenc

Phase:- The position of a point in time (instant) on a wave form cycle.

Calculation:

Consider a square pulse ——

$$x(t) = \begin{cases} 1, & -T_0 < t < T_0 \\ 0, & |t| > 0 \end{cases}$$



We know that, the equation of the Fourier Transform,

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$= \int_{-T_0}^{T_0} 1 \cdot e^{-j\omega t}\, dt \quad [\because |t| > 0]$$

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$$= \int_{-T_0}^{T_0} e^{-j\omega t}$$

$$= \left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-T_0}^{T_0}$$

$$= \left[\frac{e^{-j\omega T_0} - e^{+j\omega T_0}}{-j\omega}\right]$$

$$= \left[\frac{-(e^{j\omega T_0} - e^{-j\omega T_0})}{+j\omega}\right]$$

$$= \frac{2}{\omega}\left[\frac{e^{j\omega T_0} - e^{-j\omega T_0}}{2j}\right]$$

$$= \frac{2}{\omega} \sin(\omega T_0)$$

$$= 2T_0 \frac{\sin(\omega T_0)}{\omega T_0}$$

$$= 2T_0 \operatorname{sinc}(\omega T_0)$$

From Fourier transform of this signal, we have,

$X(j\omega) = 2T_0 \, Sinc(\omega T_0)$

To compare the signal with $A \, Sinc(\pi)$, we get

The amplitude of the signal, $A = 2T_0$

and $\omega T_0 = \pi$

$\therefore \omega = \frac{\pi}{T_0}$

We consider the time period of the signal, $T_0 = 2$, rad

Amplitude, $A = 2T_0 = 2 \times 2 = 4$.

and $\omega = \frac{\pi}{T_0} = \frac{\pi}{2}$ Per cycle.

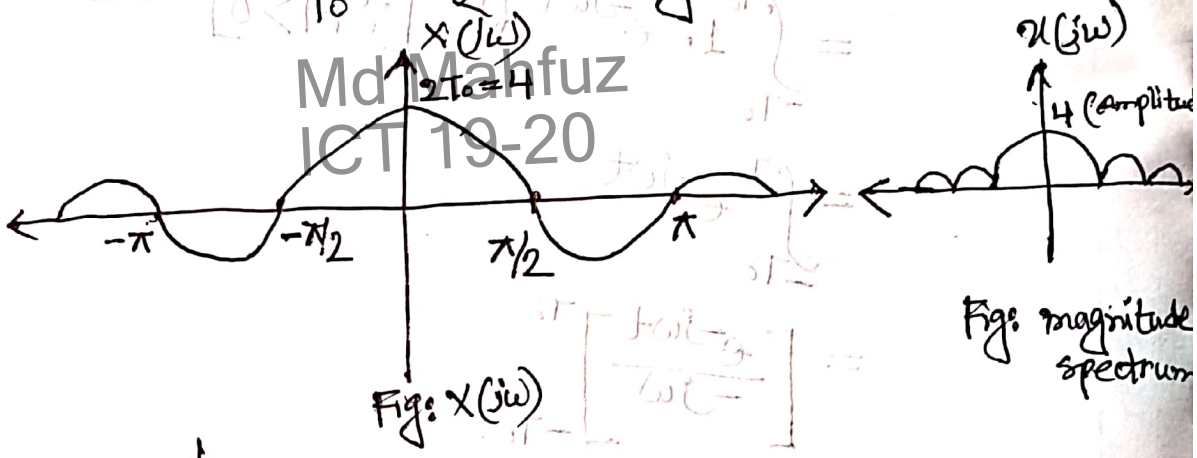

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Fig: magnitude spectrum

Fig: $X(j\omega)$

Phase spectrum

$$\angle X(j\omega) = \theta(\omega) = \begin{cases} 0 & \text{for } \omega = 0 \\ \pi & Sinc(\omega T_0) < 0 \text{ and } \omega < 0 \\ -\pi & Sinc(\omega T_0) > 0 \text{ and } \omega \end{cases}$$

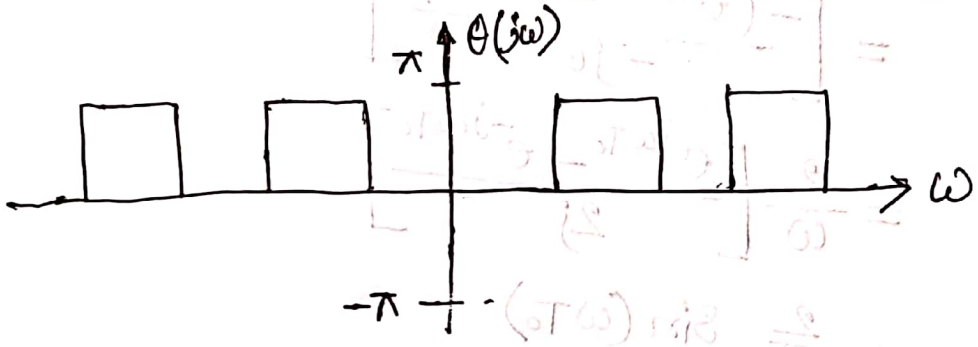

Fig: phase spectrum.

Algorithm:

(i) Start
(ii) Input the signal of $x(t)$ and desired time (t).
(iii) Define $x(t) \Rightarrow (t>=1) - (t>=1)$.
(iv) Plot the input signal.
(v) Determine the Fourier transform of the signal using fft function. or manually $X = 2T_0 \text{sinc}(\omega T_0)$
(vi) To determine the amplitude of the signal use abs() library function for magnitude spectrum.
(vii) The higest of of the magnitude spectrum is amplitude and plot it
(viii) Use angle() library function to determine the phase spectrum and plot the result.
(ix) Stop.

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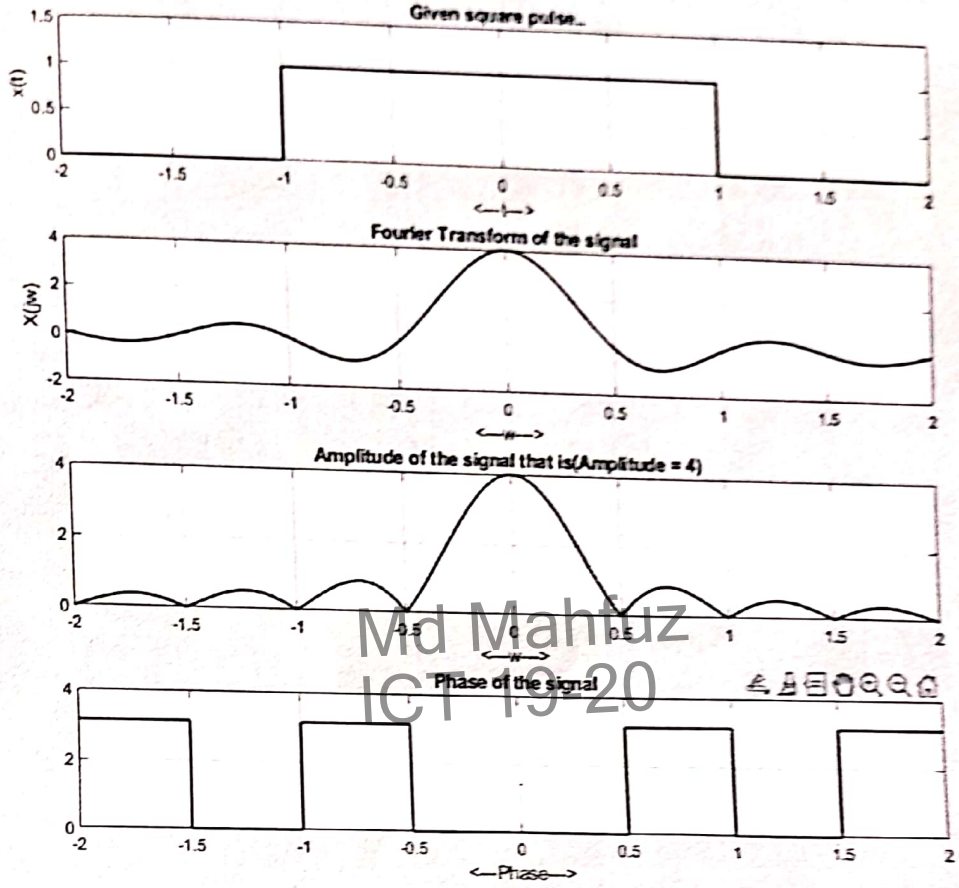
Result & Disscussion:

From the calculation, we get, amplitude is = 4.

and phase, $\theta(j\omega) = \begin{cases} 0 & \text{for } \omega = 0 \\ +\pi & \text{sinc}(\omega T_0) < 0 \text{ and } \omega < 0 \\ -\pi & \text{sinc}(\omega T_0) > 0 \text{ and } \omega > 0 \end{cases}$

Output:



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Experiment No: 10

Name of the experiment: Generate a discrete time sequence by sampling a continuous time signal. Show that with sampling rates less than Nyquist rate, aliasing occurs while reconstructing the signal.

AIM:

To find alising occurs when the ~~Ny~~ samples rates less than Nyquist rate and reconstruct the signal.

Objectives:

(i) Know about how to convert continuous time signal
(ii) To learn about under sampling and Nyquist rate.
(iii) To observe signal after sampling and reconstructing using MAT-LAB.
(iv) To know about aliasing effect.

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Theory:

Sampling: Sampling is the process to convert continuous time signal into discrete time signal. In signal processing, sampling is the reduction of continuous time signal to a discrete time signal.

If a signal band limited, that means its fourier transform is zero outside a finite band of frequencies, and if the samples are taken sufficiently close together in relation highest frequency present in the signal, thus we can reconstruct it perfectly.

Consider a continuous time signal $x(t)$. The spectru of $x(t)$ is band limited to ω_m Hz that is, the spectrum of $x(t)$.

Sampling of input signal $x(t)$ can be obtained by multiplying $x(t)$ with an impulse train $\delta(t)$ of period T_s. Shown the following diagram is the below:

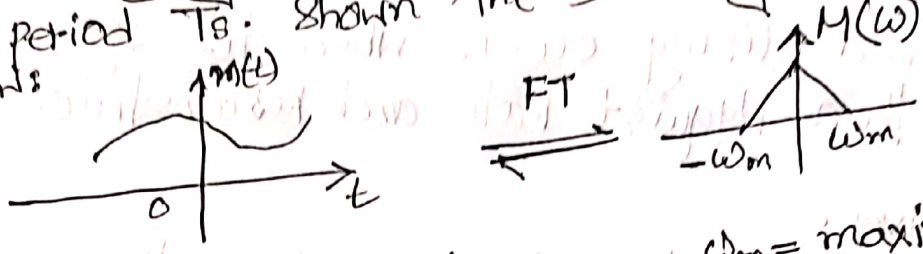

Fig: Message signal

ω_m = maximum freque component.

The output of multiplier is discrete sample signal is represented with $y(t)$ in the following diagra

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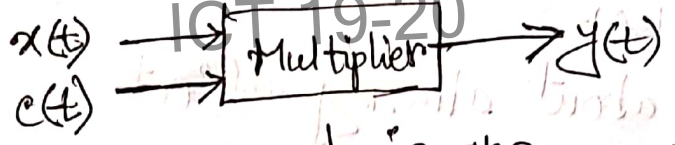
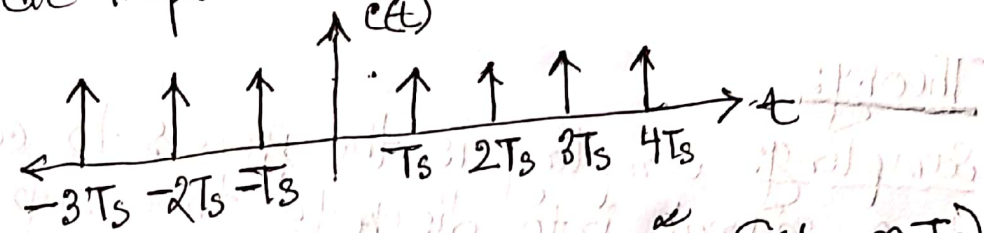
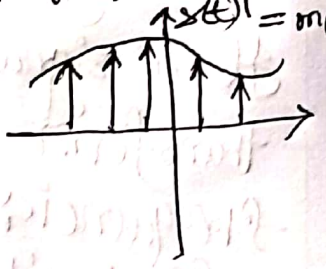


The periodic impulse train are ——



and the impulse train, $c(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$

Now, obtaining the spectrum of the sampled signal, we have,

$s(t) \rightleftharpoons S(\omega)$ and $s(t) =$ 

$\therefore S(t) = m(t) \cdot c(t)$

$\Downarrow$ FOT

$$S(\omega) = \frac{1}{2\pi} [M(\omega) * C(\omega)]$$

Here, $C(\omega) = \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$

$$\therefore S(\omega) = \frac{1}{2\pi}\left[M(\omega) * \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)\right]$$

$$= \frac{\omega_s}{2\pi}\left[M(\omega) * \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)\right]$$

$$= \frac{1}{T_s}\left[\sum_{n=-\infty}^{\infty} M(\omega) * \delta(\omega - n\omega_s)\right]$$

$$= \frac{1}{T_s}\left[\sum_{n=-\infty}^{\infty} M(\omega - n\omega_s)\right] \quad [\because x(t) * \delta(t-t_1) = x(t-t_1)$$

$$= \frac{1}{T_s}\left[---- + M(\omega+\omega_s) + M(\omega) + M(\omega-\omega_s) + ---\right]$$

Using $S(\omega)$ expansion, we shown that the ~~sor~~ classification of sampling:

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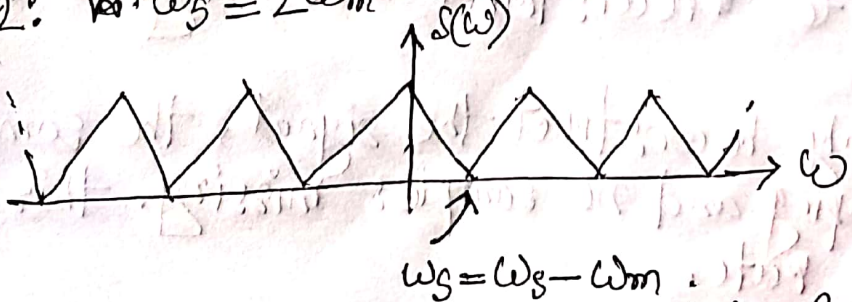
Under sampling (Aliasing effect occurs):

Case 1: $\omega_s < 2\omega_m$

In the following figure, the triangle are over lapping, so some sample are missing to reconstructing the original signal, ~~thing~~ this is called aliasing. This process is under sampling process.

Nyquist rate: (Perfect sampling)

Case 2: $\omega_s = 2\omega_m$



In this figure, no sample loses, that is perfect for sampling.

Calculation:

Let us consider a continuous time signal which has 5 Hz frequency $\longrightarrow f_m = 5Hz$

$\therefore x(t) = \sin(2\pi f_m t) = \sin(2\pi \times 5 \times t)$

When sampling rates less than Nyquist rate:

If sampling rates less than Nyquist rate, this is called under sampling.

Under sampling:

From the experiment theory, we know that —

$\omega_s < 2\omega_m$, the under sampling occurs.

$\Rightarrow 2\pi f_s < 2 \cdot 2\pi f_m$

$\Rightarrow f_s < 2f_m$.

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Let $f_s = 5$ that $5 < 2 \cdot 5 \Rightarrow 5 < 10$, the under sampling

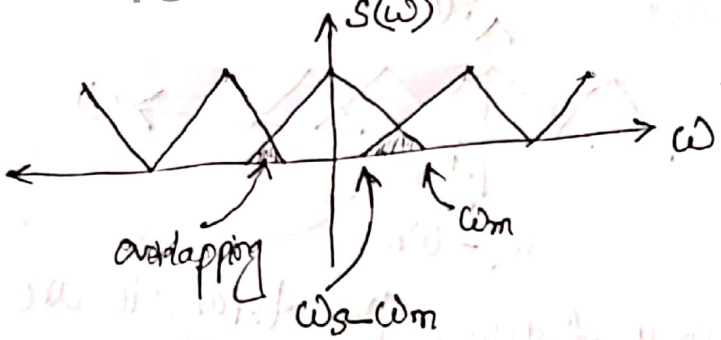

Figure: Under sampling

In the following figure, the samples are overlapping to each other, so the overlapped signal are missing or loses in that time to reconstruct the main signal. The loses or missing samples occurs the aliasing to reconstruct the original signal.

To properly reconstruct the signal, the samples are no overlaping and no samples missing. this is the Nyquist rate.

When sampling rates is equal to Nyquist Rate:

Nyquist rate:

From the experiment theory, we know that

$\omega_s = 2\omega_m$

$\Rightarrow 2\pi f_s = 2 \times 2\pi f_m$

$\Rightarrow f_s = 2 f_m$

$= 2 \times 5 = 10$ Hz

When $f_s = 10$ Hz, then the $\omega_s = 2\omega_m$ condition is full. Now, the following figure into the condition.

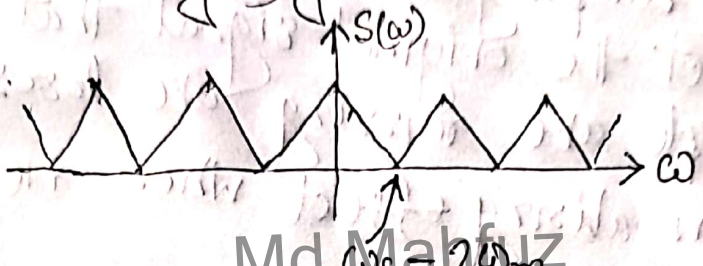

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Figure: Nyquist rate (perfect sampling)

In the following figure, there are no overlapping, and the samples just touch each other. That is perfect for sampling. Using Nyquist Rate, we plot the discrete time signal (converted)

Algorithm:

(i) Start
(ii) Input the continuous time signal with $f_m = 5$ Hz.
(iii) Plot the continuous time signal.
(iv) Work with under sampling process that $f_s < 2f_m$
(v) Reconstruct the signal with under sampling process and show that aliasing effect
(vi) Work with Nyquist rate that $f_s = 2f_m$.
(vii) Reconstruct the signal with Nyquist rate and stem to plot the discrete time signal.
(viii) Stop.

Result and Discussion:

From the calculating and plotting figure,
Only Nyquist rate, that $f_s = 2f_m$, we can reconstruct the original signal perfectly.
We are also convert discrete time sampled si using the Nyquist rate.

On the otherhand,
In under sampling rate, $f_s < 2f_m$, we can not reconstruct the original signal that in this condi some sample are missing are loses.
So, it show aliasing effect when we reconstruct signal.

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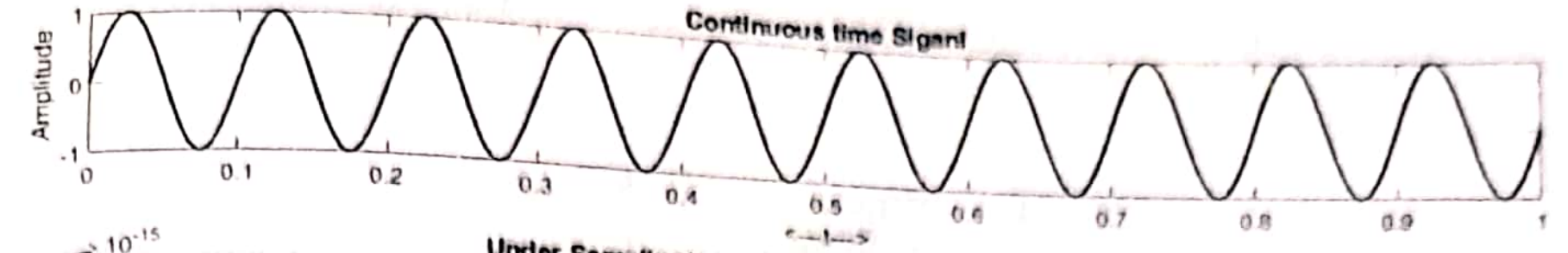

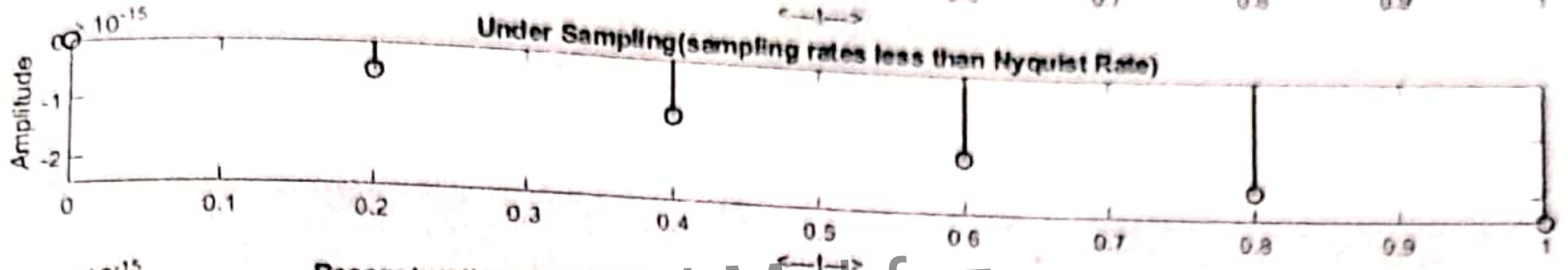

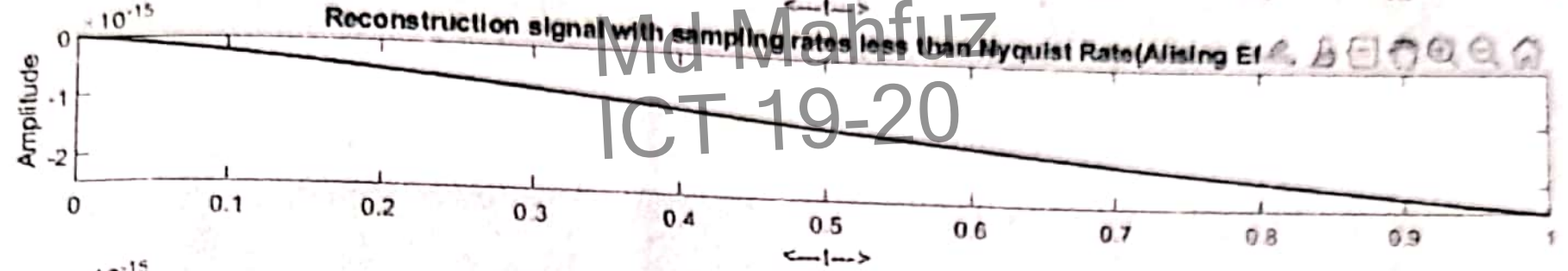

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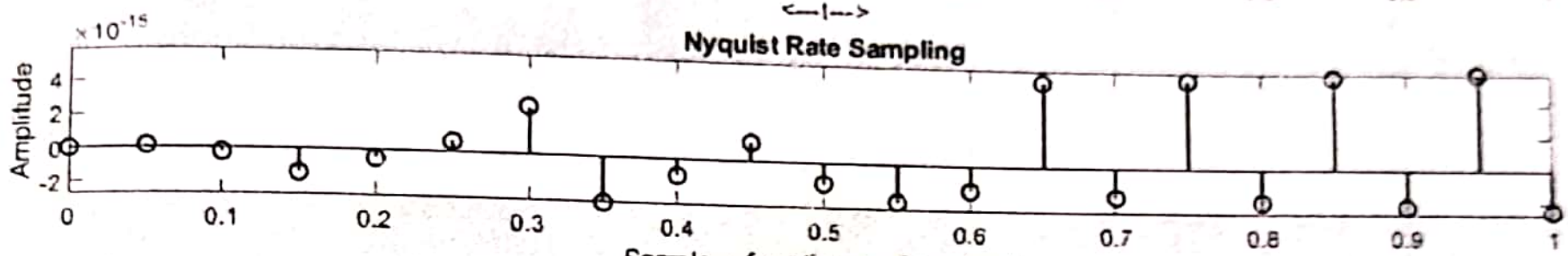

Samples of continuous time signal.